

# BFS Traversal

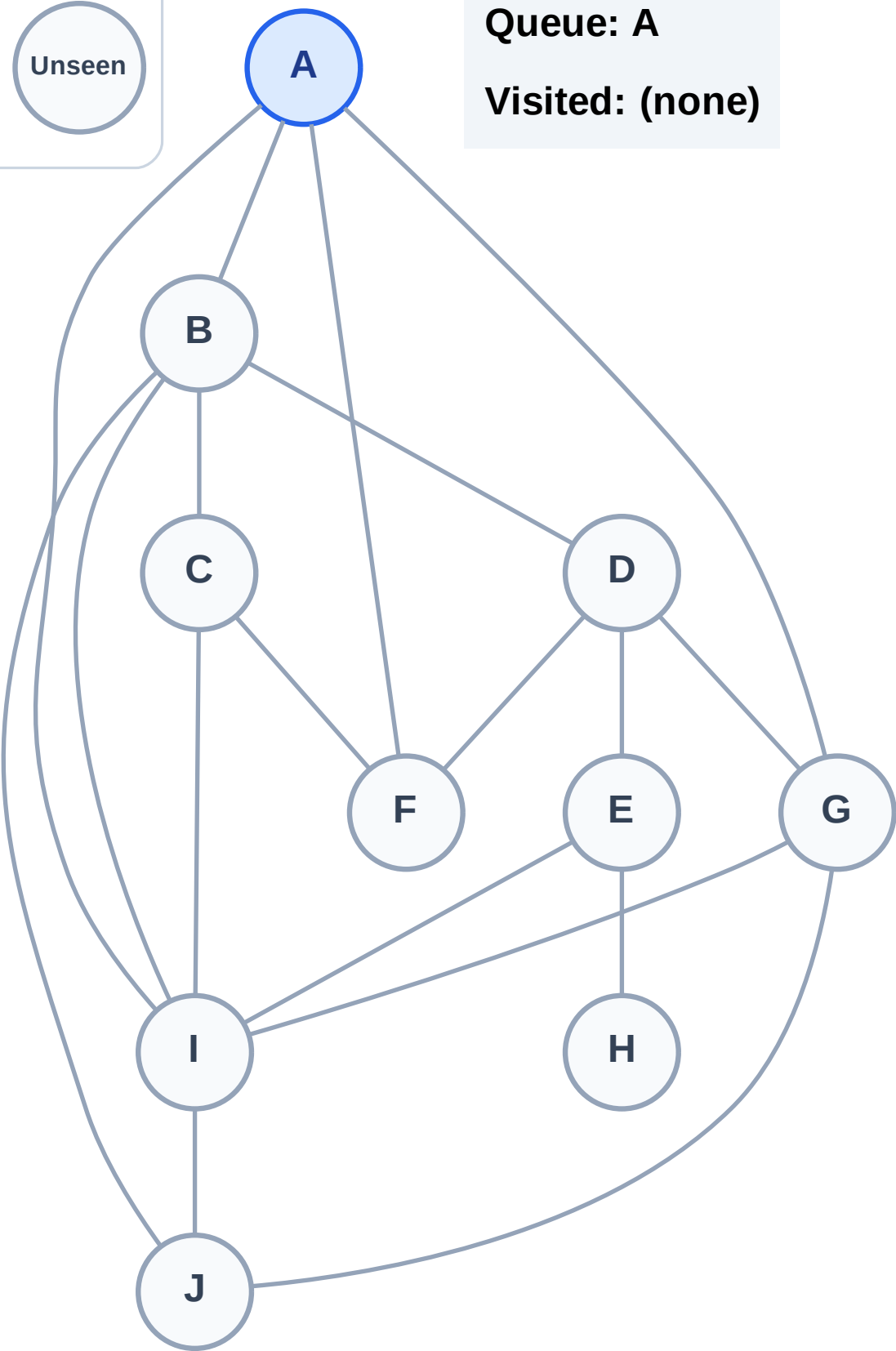
Step 1: Start at A

## Legend



Queue: A

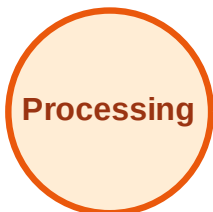
Visited: (none)



# BFS Traversal

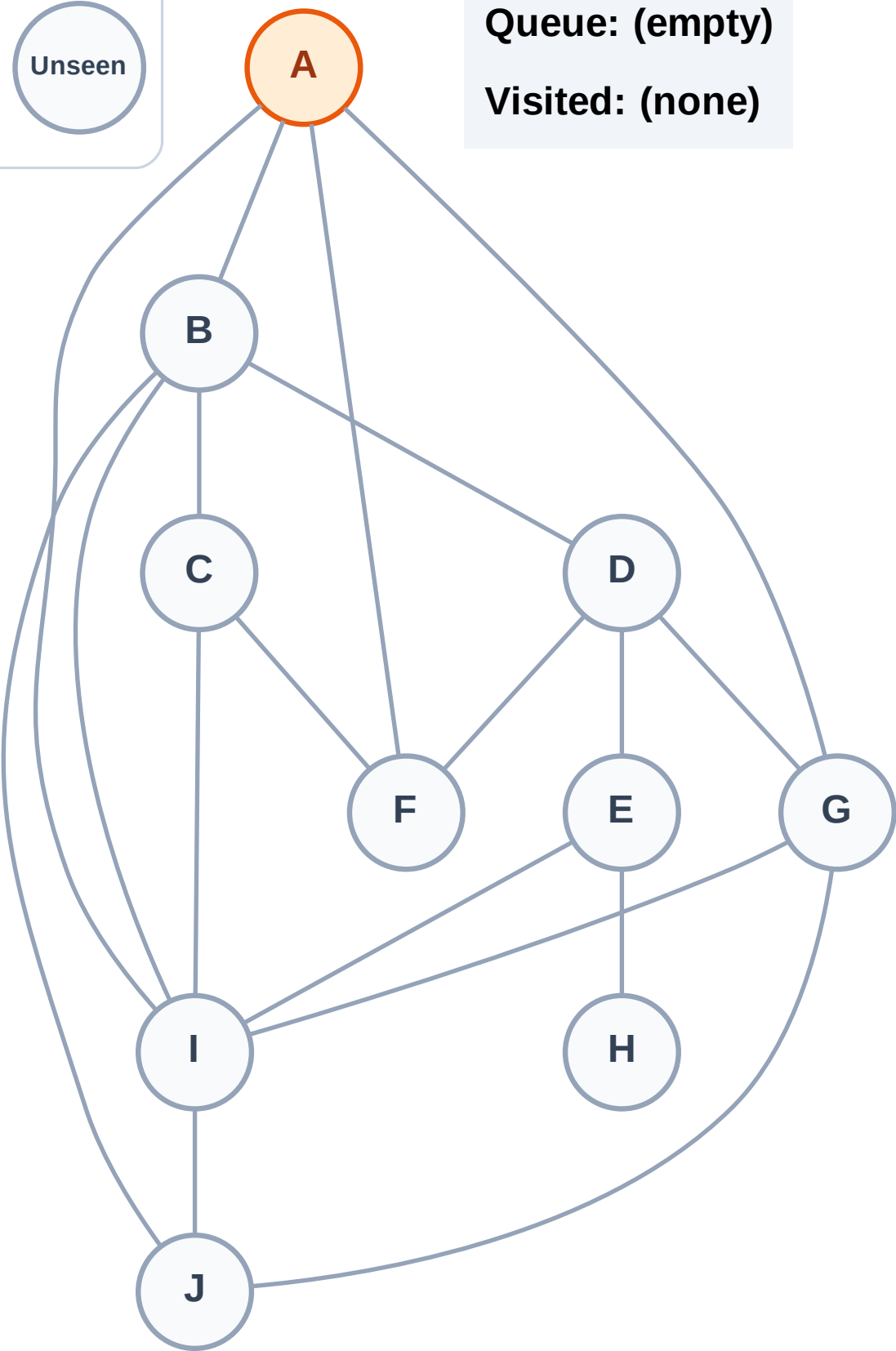
Step 2: Process node A

## Legend



Queue: (empty)

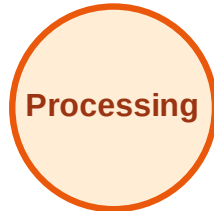
Visited: (none)



# BFS Traversal

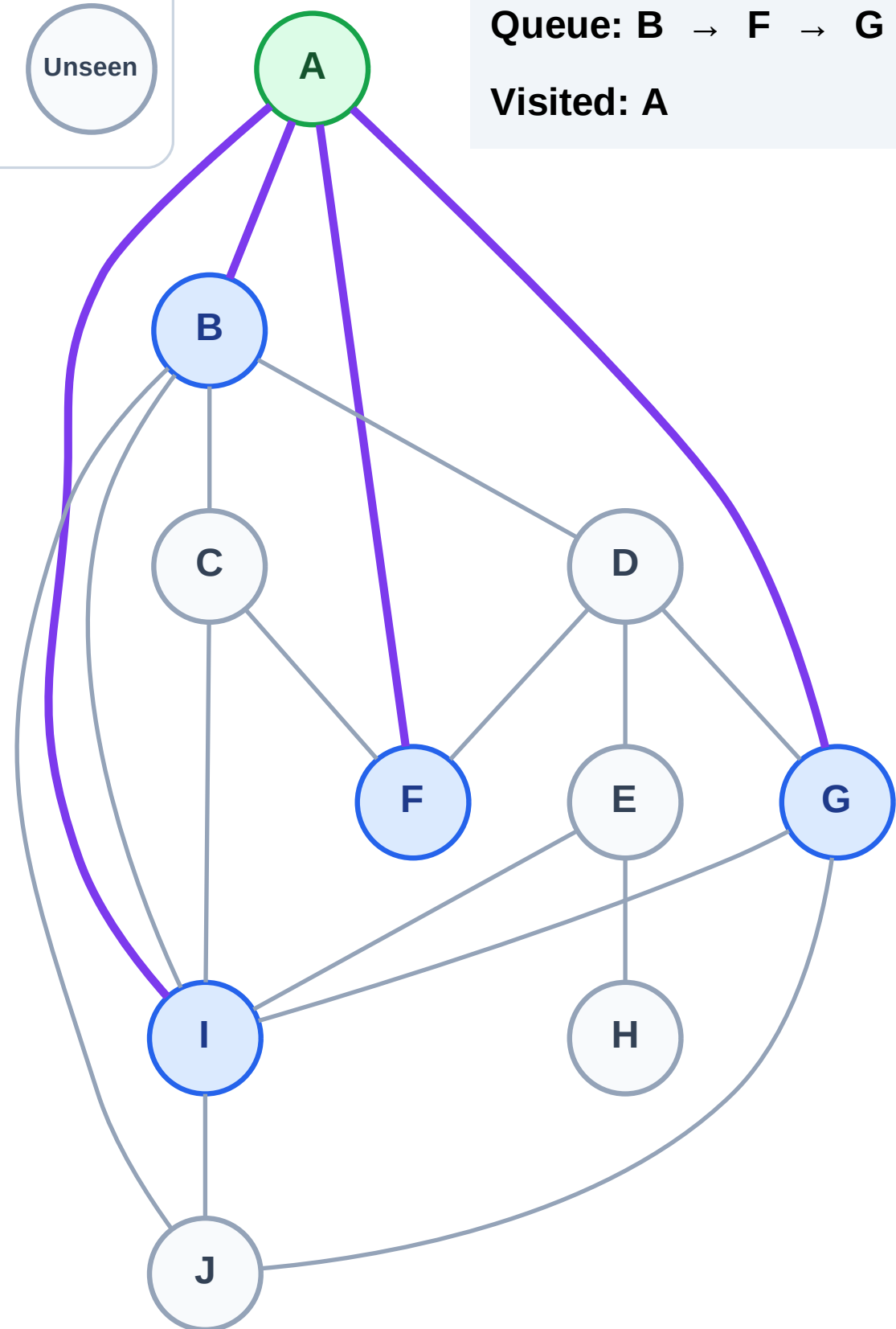
Step 3: Discover B, F, G, I from A

## Legend



Queue: B → F → G → I

Visited: A



# BFS Traversal

Step 4: Process node B

Legend

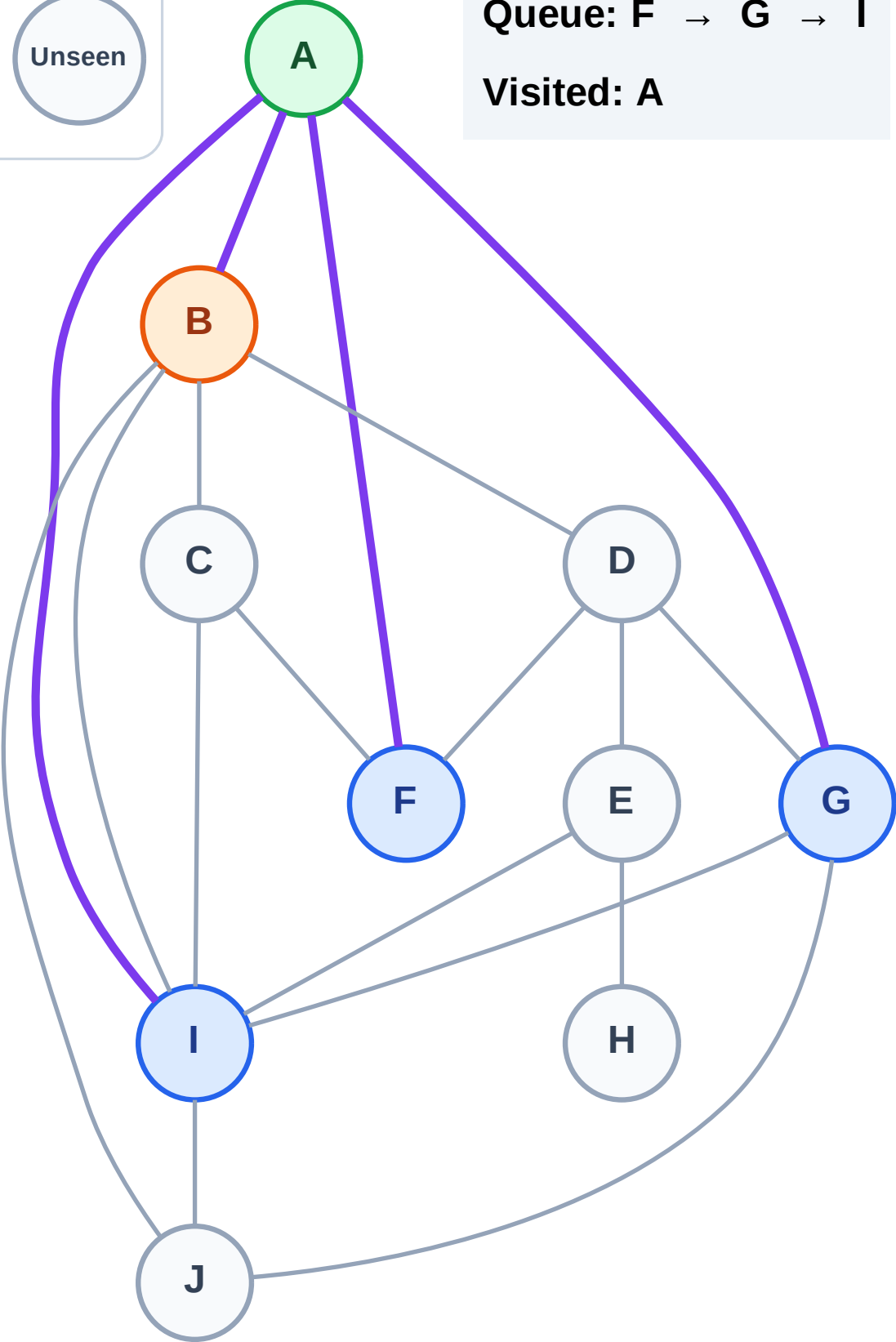
Complete

Processing

Discovered

Unseen

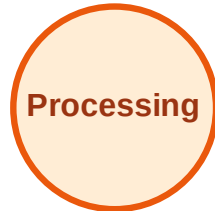
Queue: F → G → I  
Visited: A



# BFS Traversal

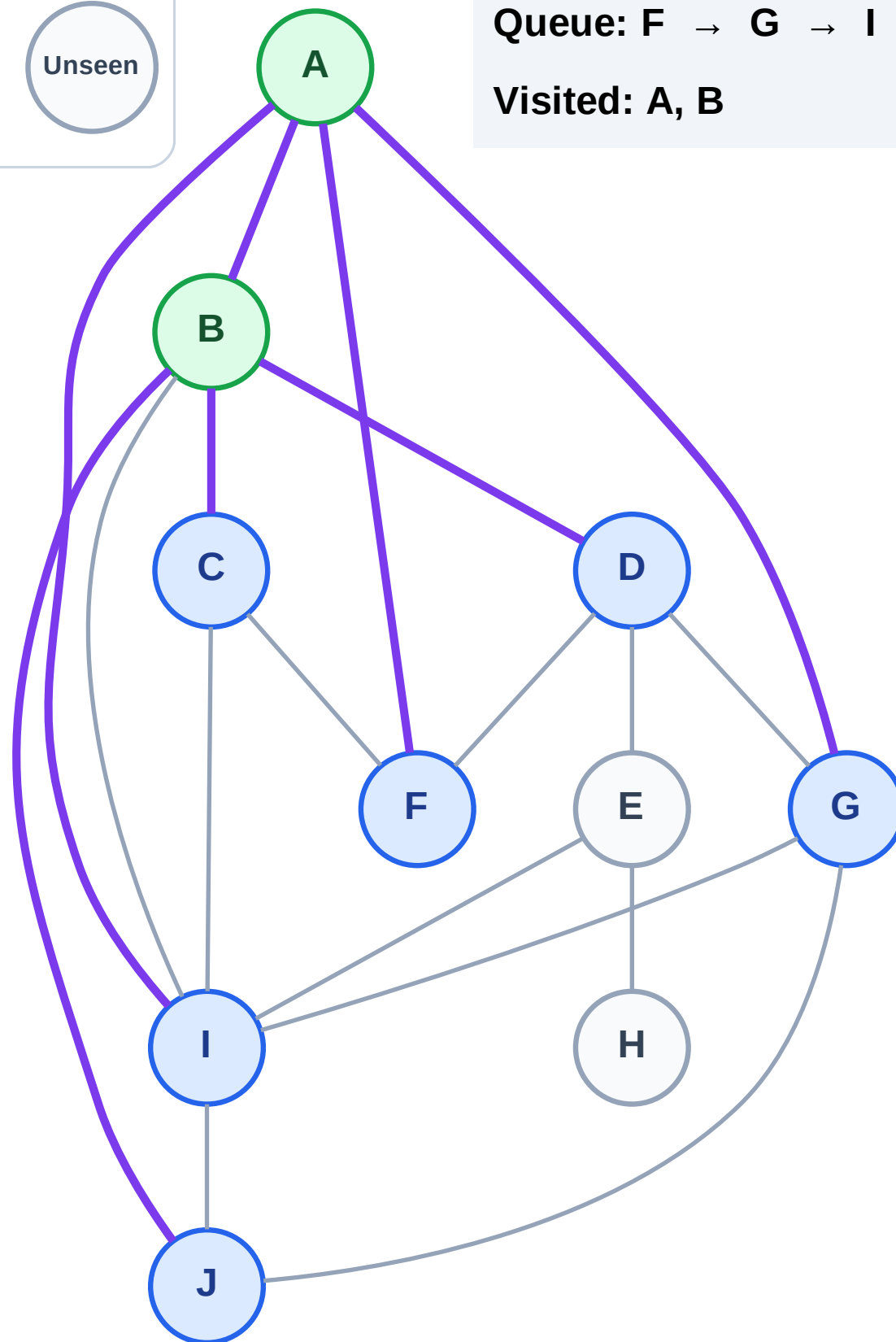
Step 5: Discover C, D, J from B

## Legend



Queue: F → G → I → C → D → J

Visited: A, B



# BFS Traversal

Step 6: Process node F

Legend

Complete

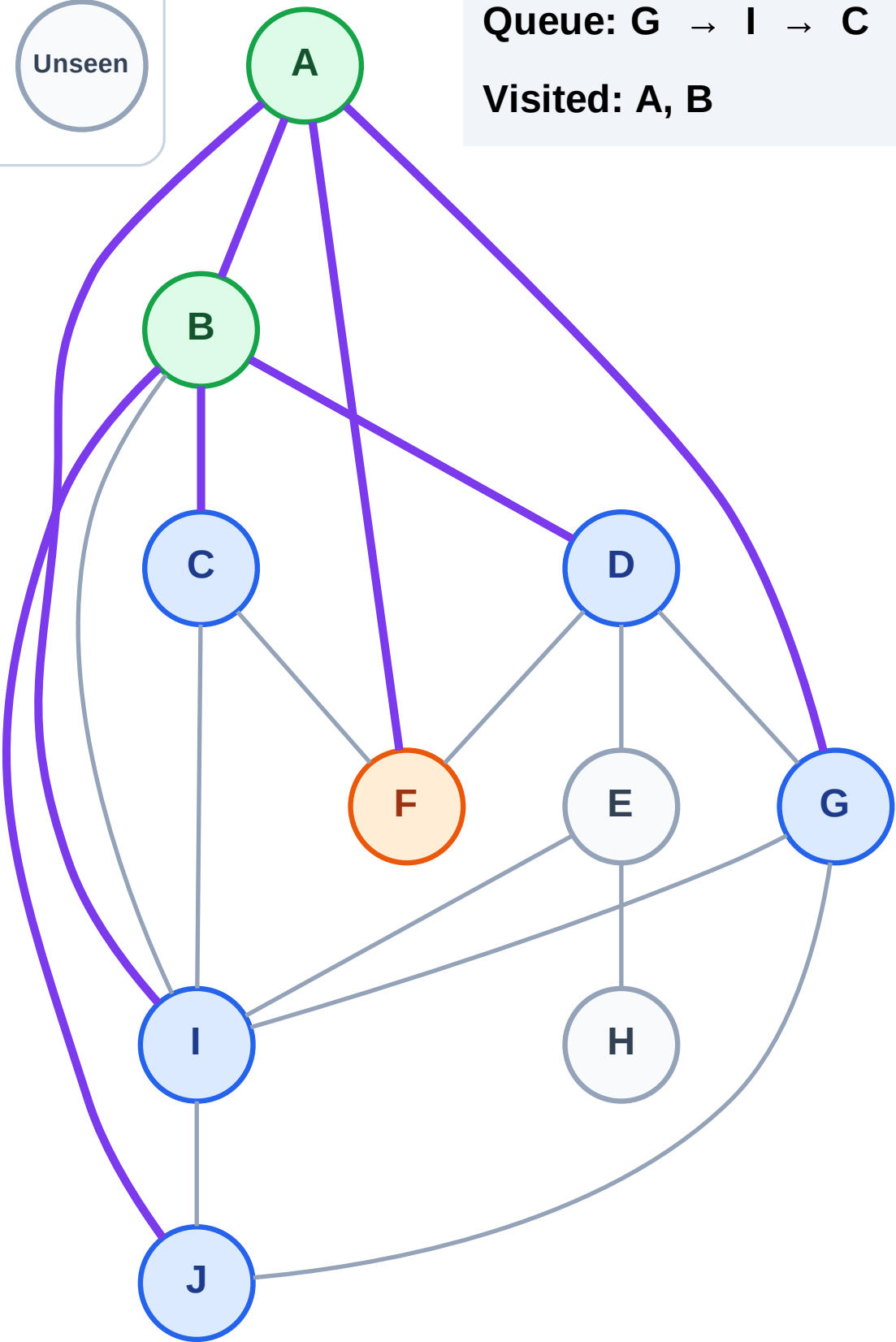
Processing

Discovered

Unseen

Queue: G → I → C → D → J

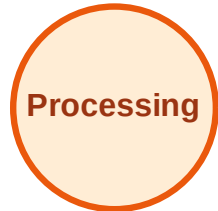
Visited: A, B



# BFS Traversal

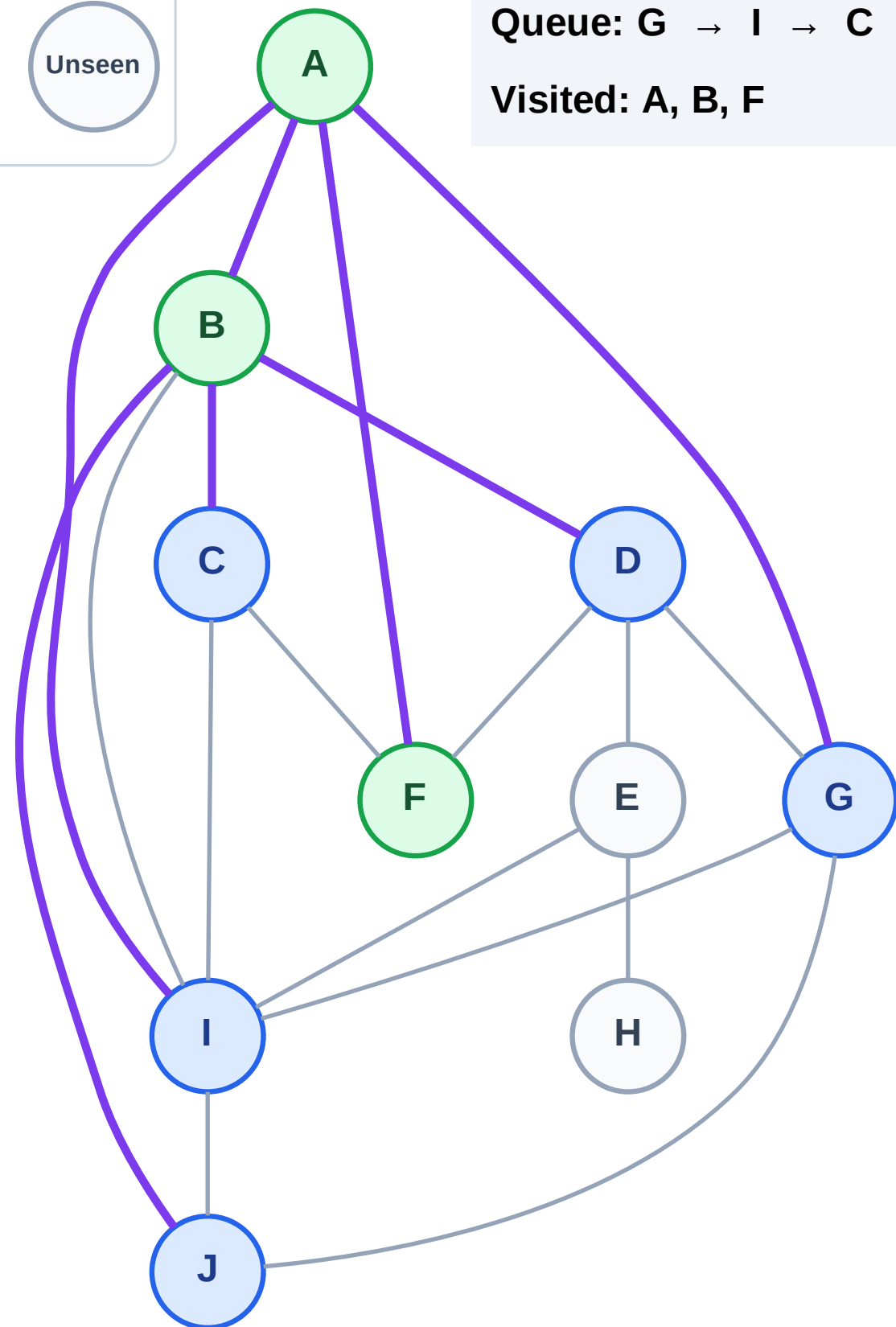
Step 7: No new neighbors from F

## Legend



Queue: G → I → C → D → J

Visited: A, B, F





# BFS Traversal

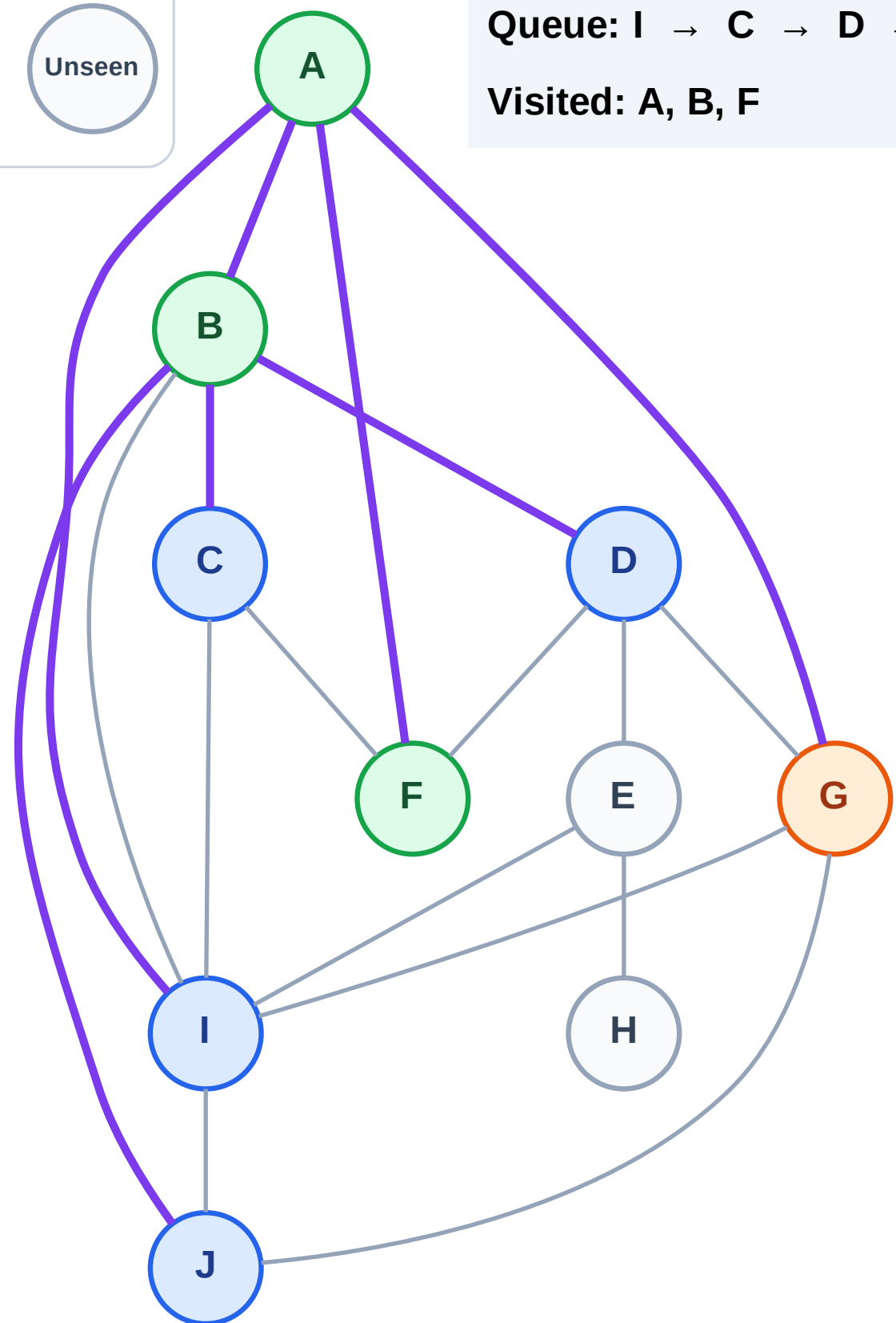
Step 8: Process node G

## Legend



Queue: I → C → D → J

Visited: A, B, F

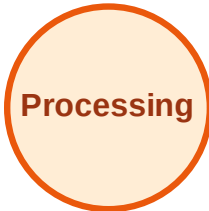




# BFS Traversal

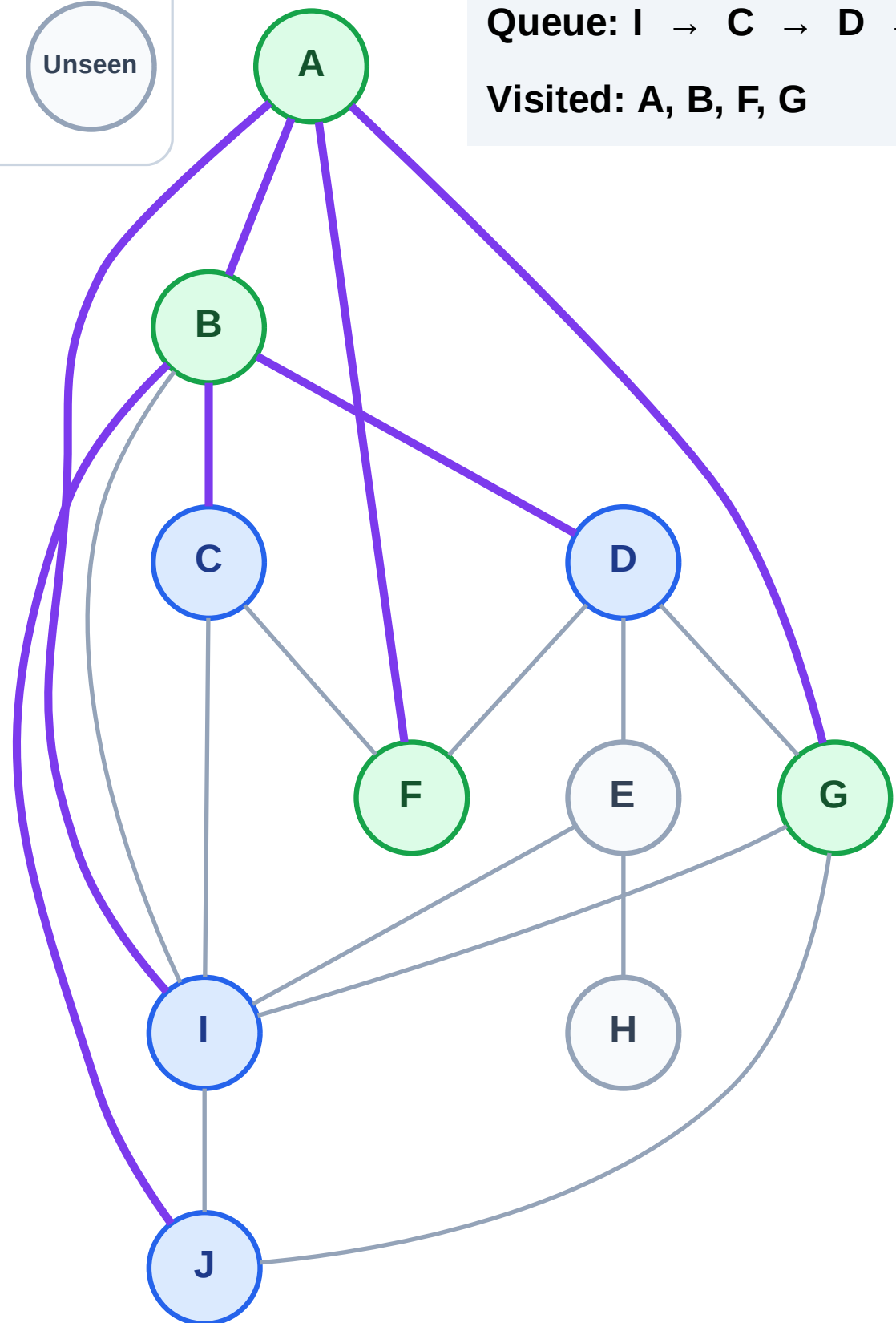
Step 9: No new neighbors from G

## Legend



Queue: I → C → D → J

Visited: A, B, F, G



# BFS Traversal

Step 10: Process node I

Legend

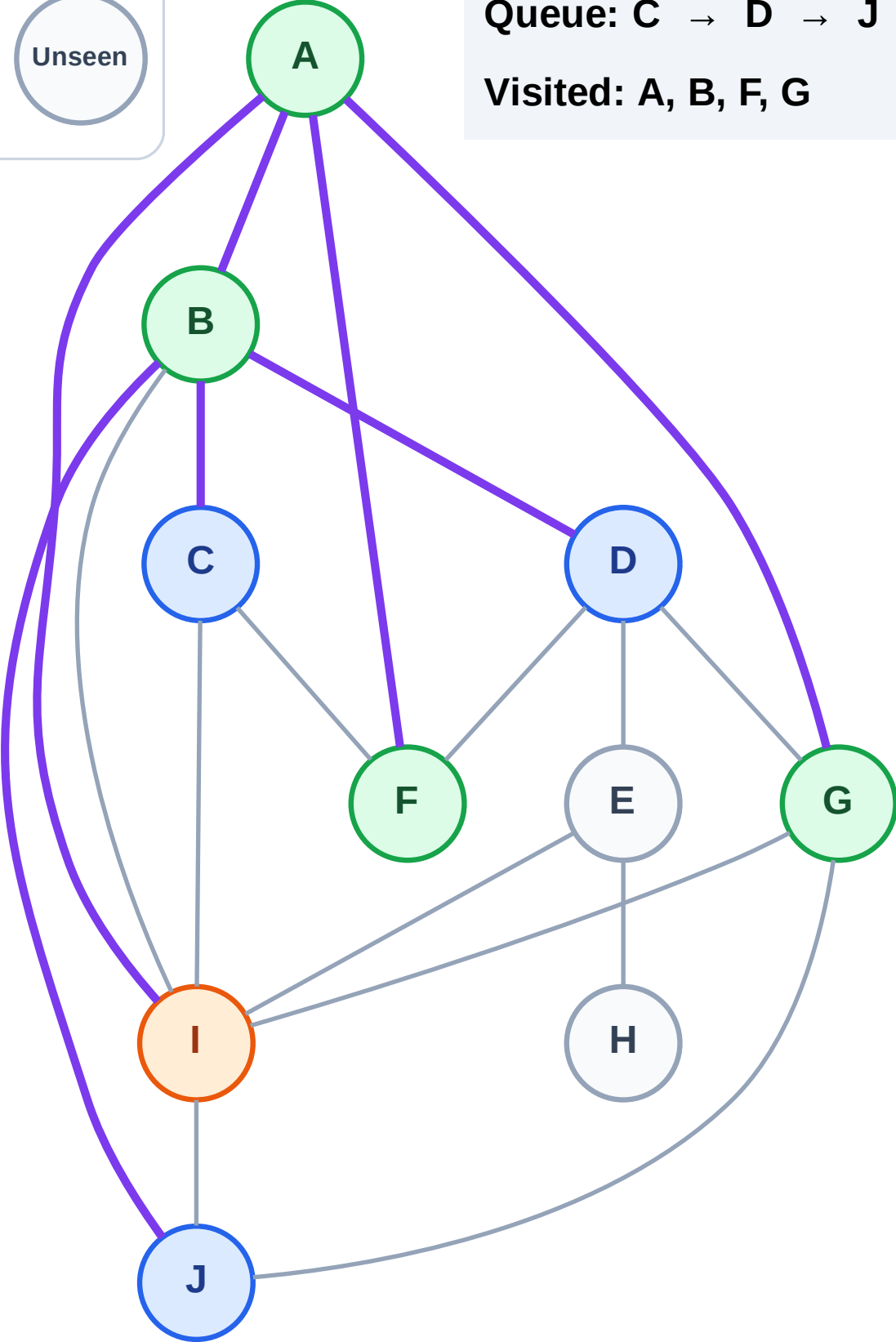
Complete

Processing

Discovered

Unseen

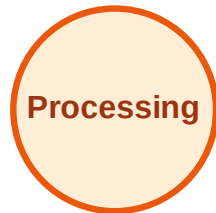
Queue: C → D → J  
Visited: A, B, F, G



# BFS Traversal

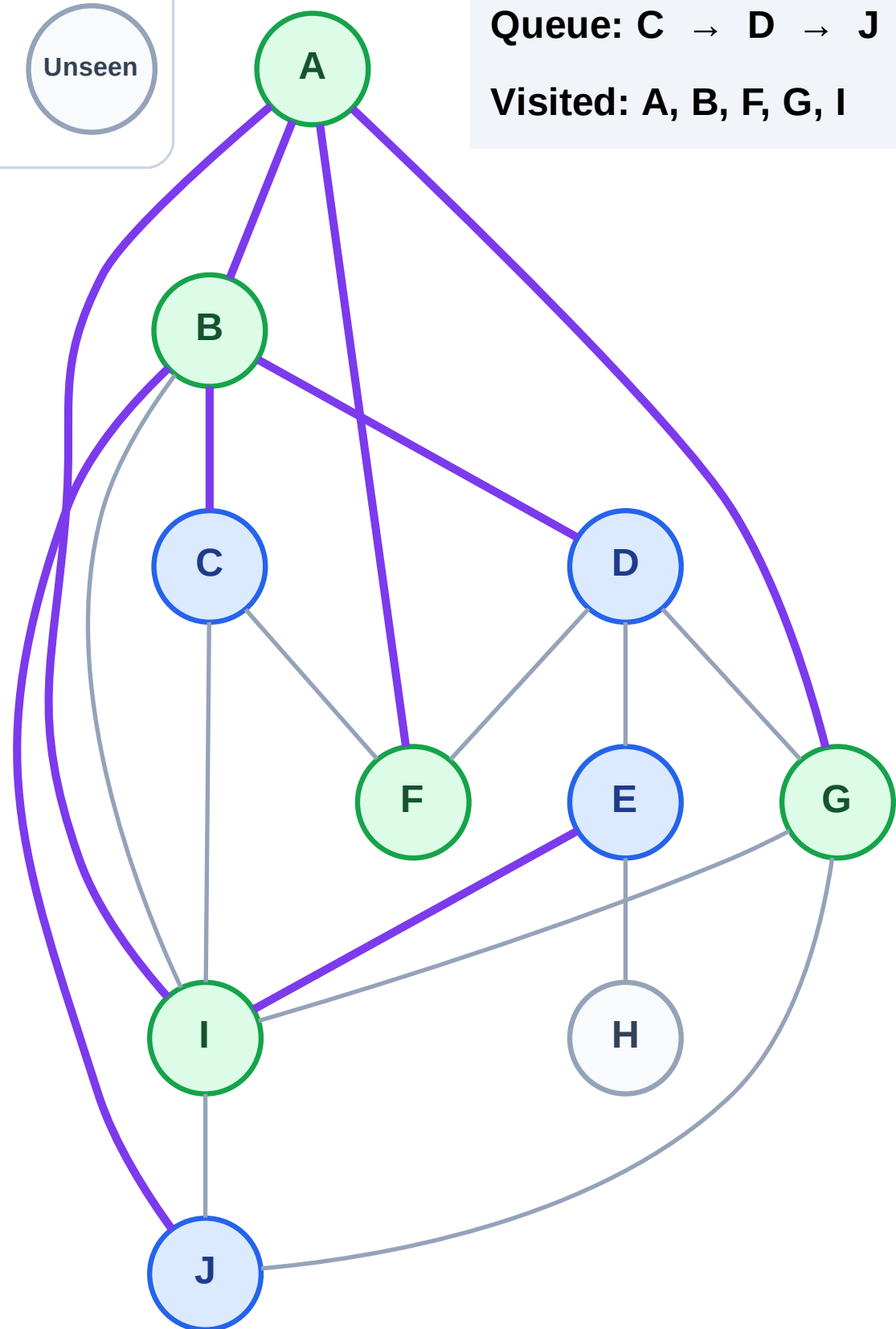
Step 11: Discover E from I

## Legend



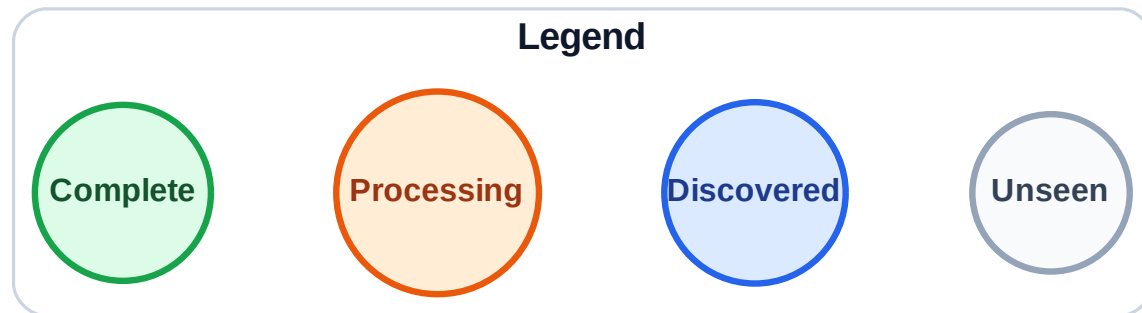
Queue: C → D → J → E

Visited: A, B, F, G, I

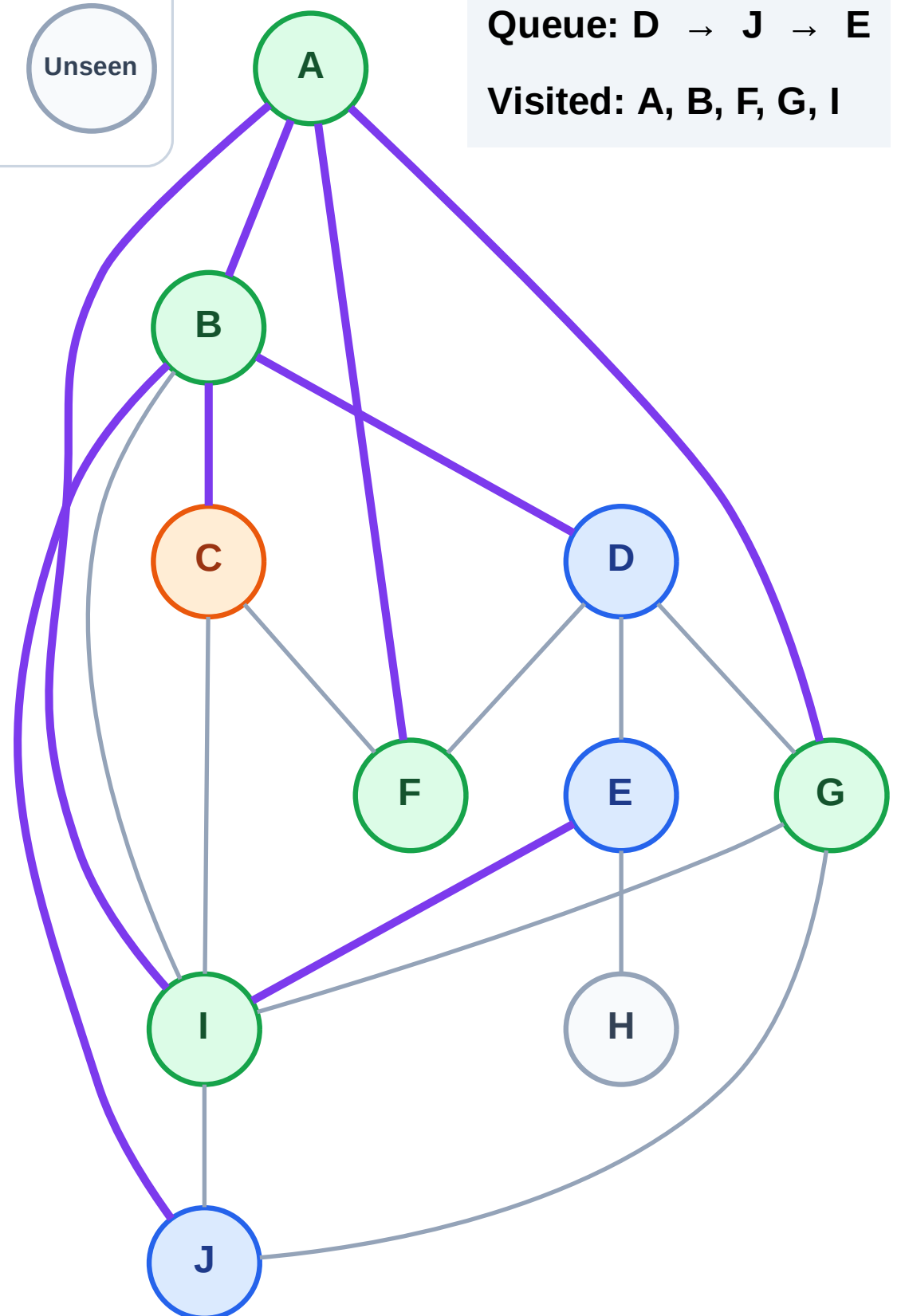


# BFS Traversal

Step 12: Process node C



Queue: D → J → E  
Visited: A, B, F, G, I



# BFS Traversal

Step 13: No new neighbors from C

Legend

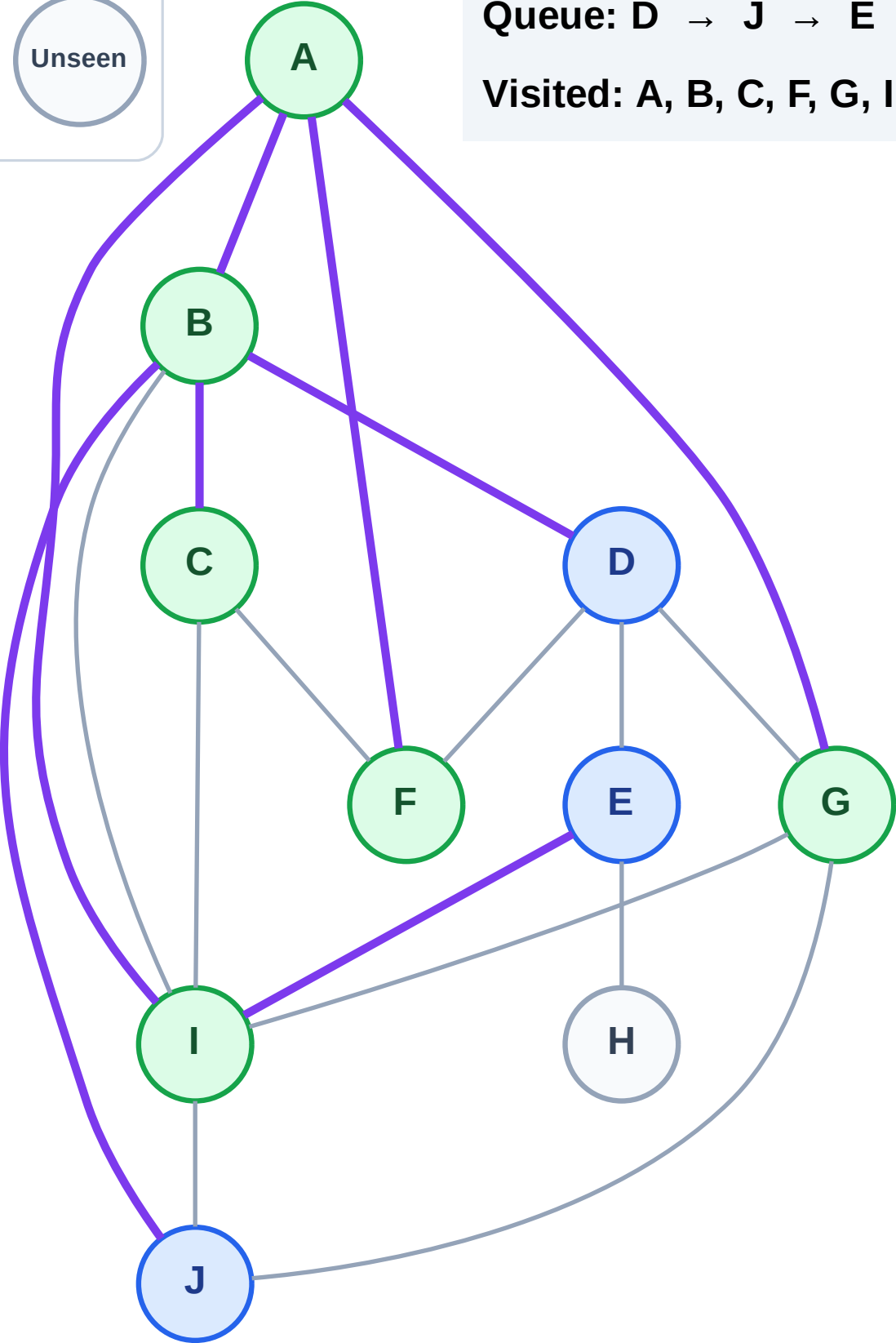
Complete

Processing

Discovered

Unseen

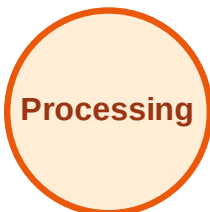
Queue: D → J → E  
Visited: A, B, C, F, G, I



# BFS Traversal

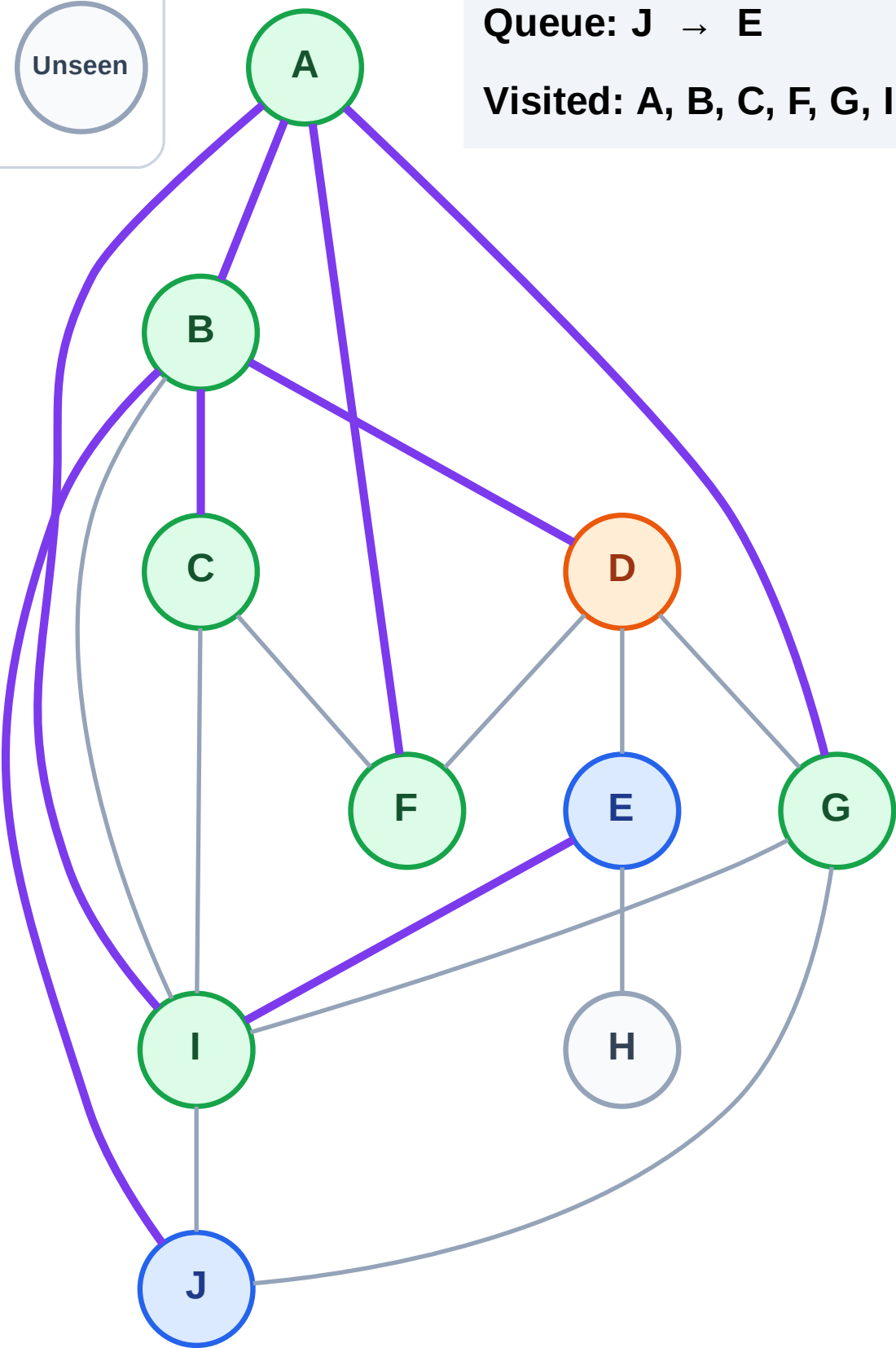
Step 14: Process node D

## Legend



Queue: J → E

Visited: A, B, C, F, G, I



# BFS Traversal

Step 15: No new neighbors from D

## Legend

Complete

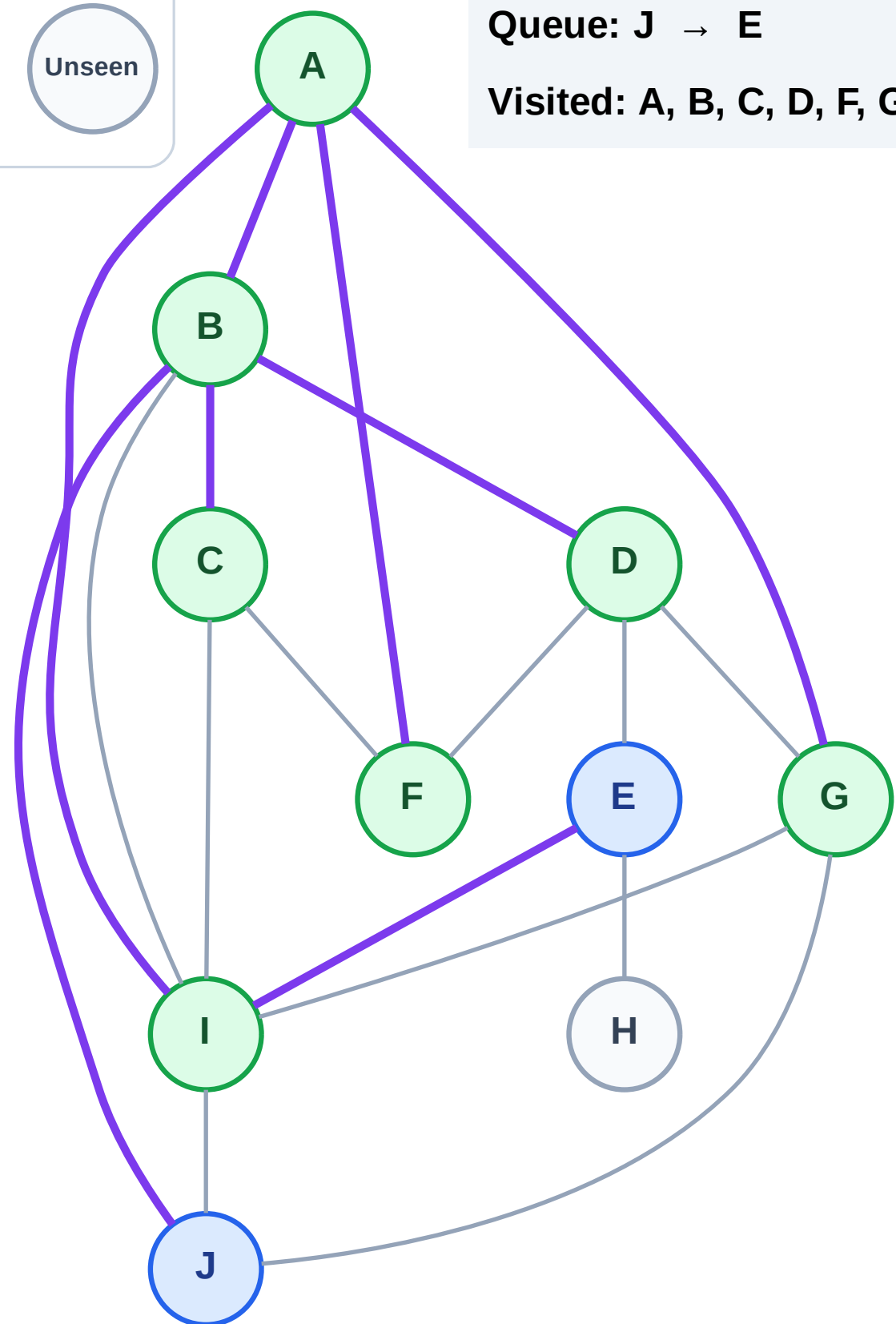
Processing

Discovered

Unseen

Queue: J → E

Visited: A, B, C, D, F, G, I



# BFS Traversal

Step 16: Process node J

Legend

Complete

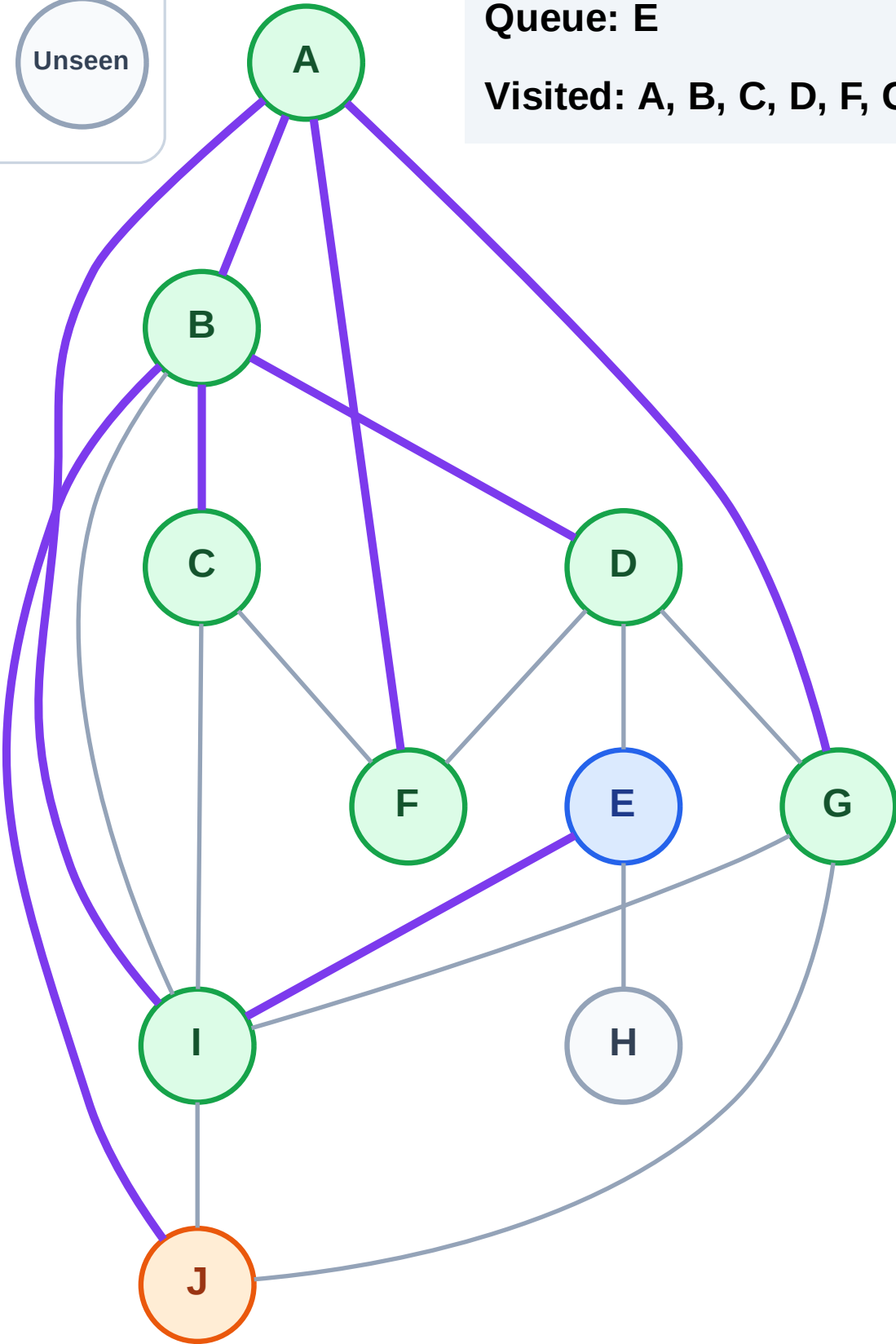
Processing

Discovered

Unseen

Queue: E

Visited: A, B, C, D, F, G, I

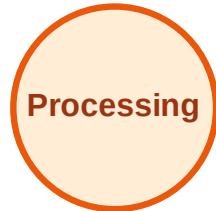




# BFS Traversal

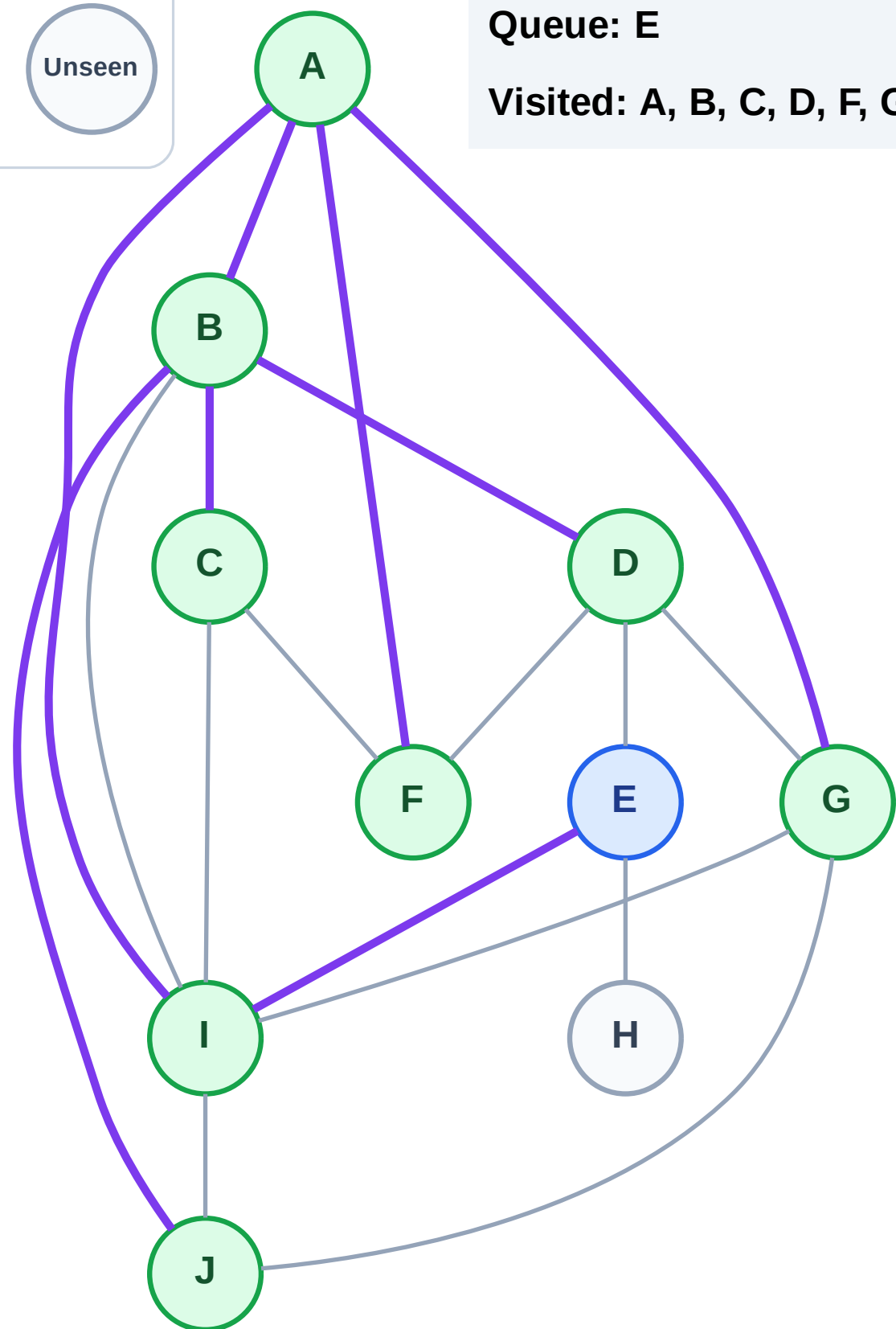
Step 17: No new neighbors from J

## Legend



Queue: E

Visited: A, B, C, D, F, G, I, J



# BFS Traversal

Step 18: Process node E

Legend

Complete

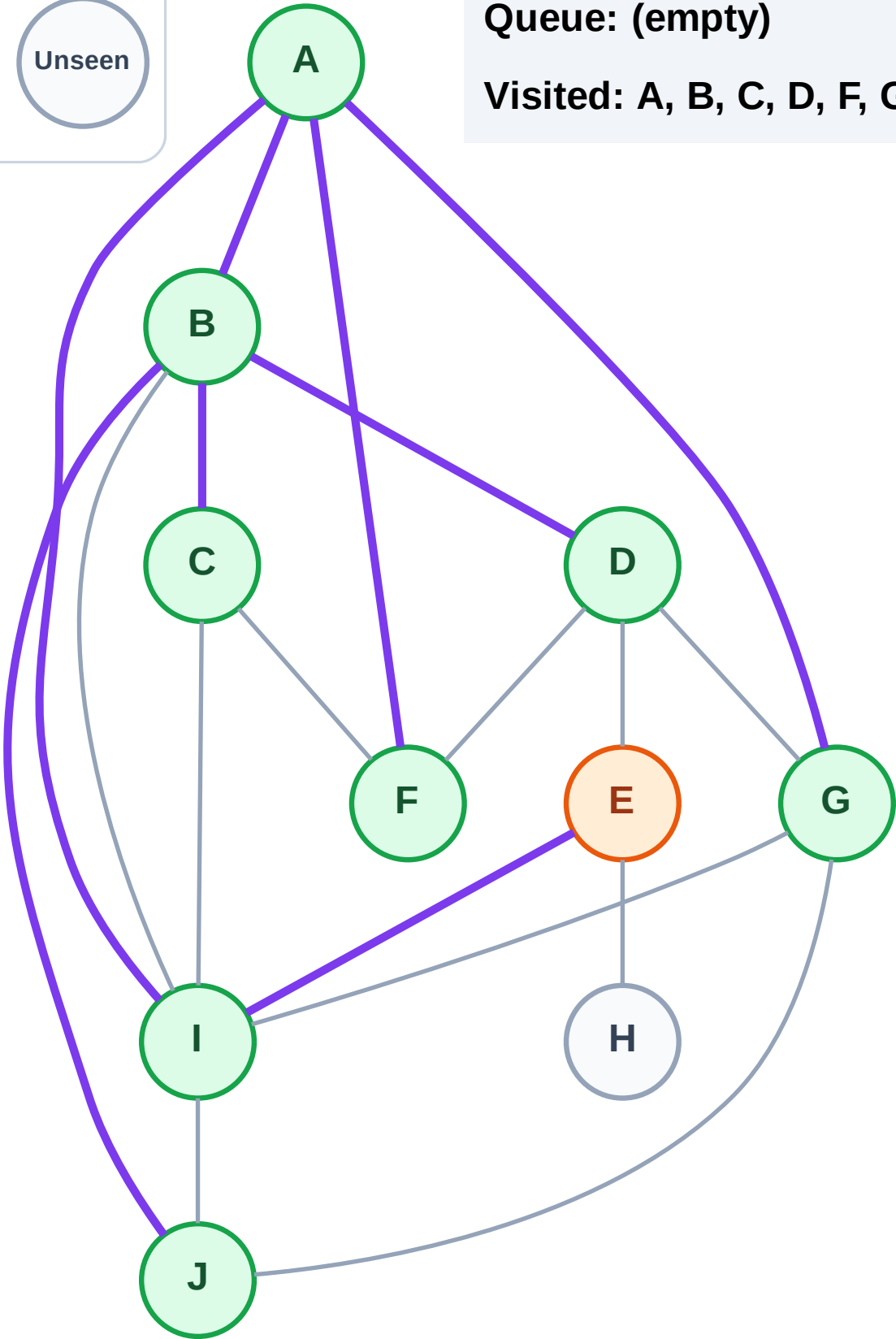
Processing

Discovered

Unseen

Queue: (empty)

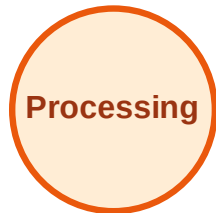
Visited: A, B, C, D, F, G, I, J



# BFS Traversal

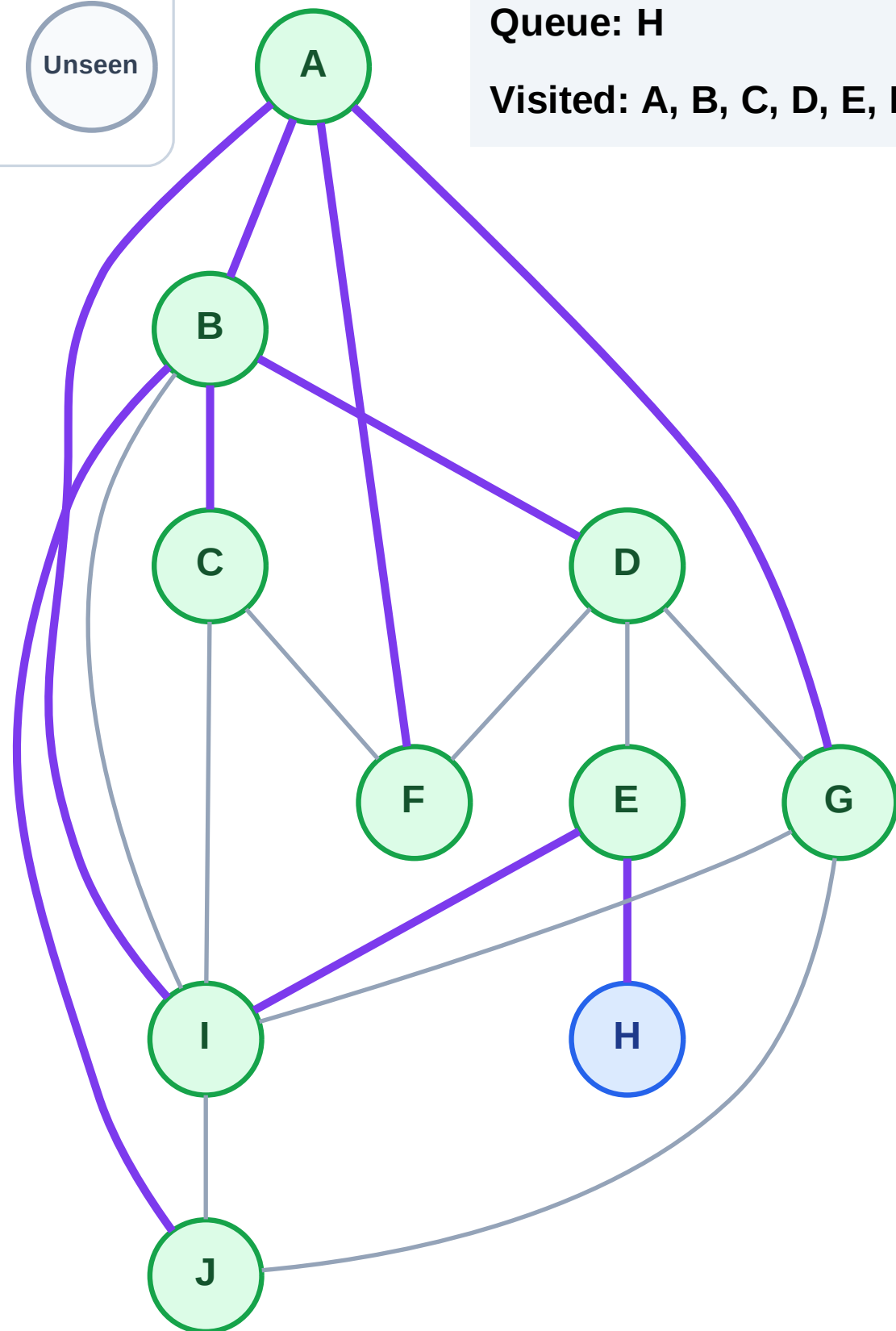
Step 19: Discover H from E

## Legend



Queue: H

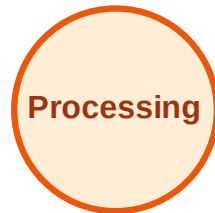
Visited: A, B, C, D, E, F, G, I, J



# BFS Traversal

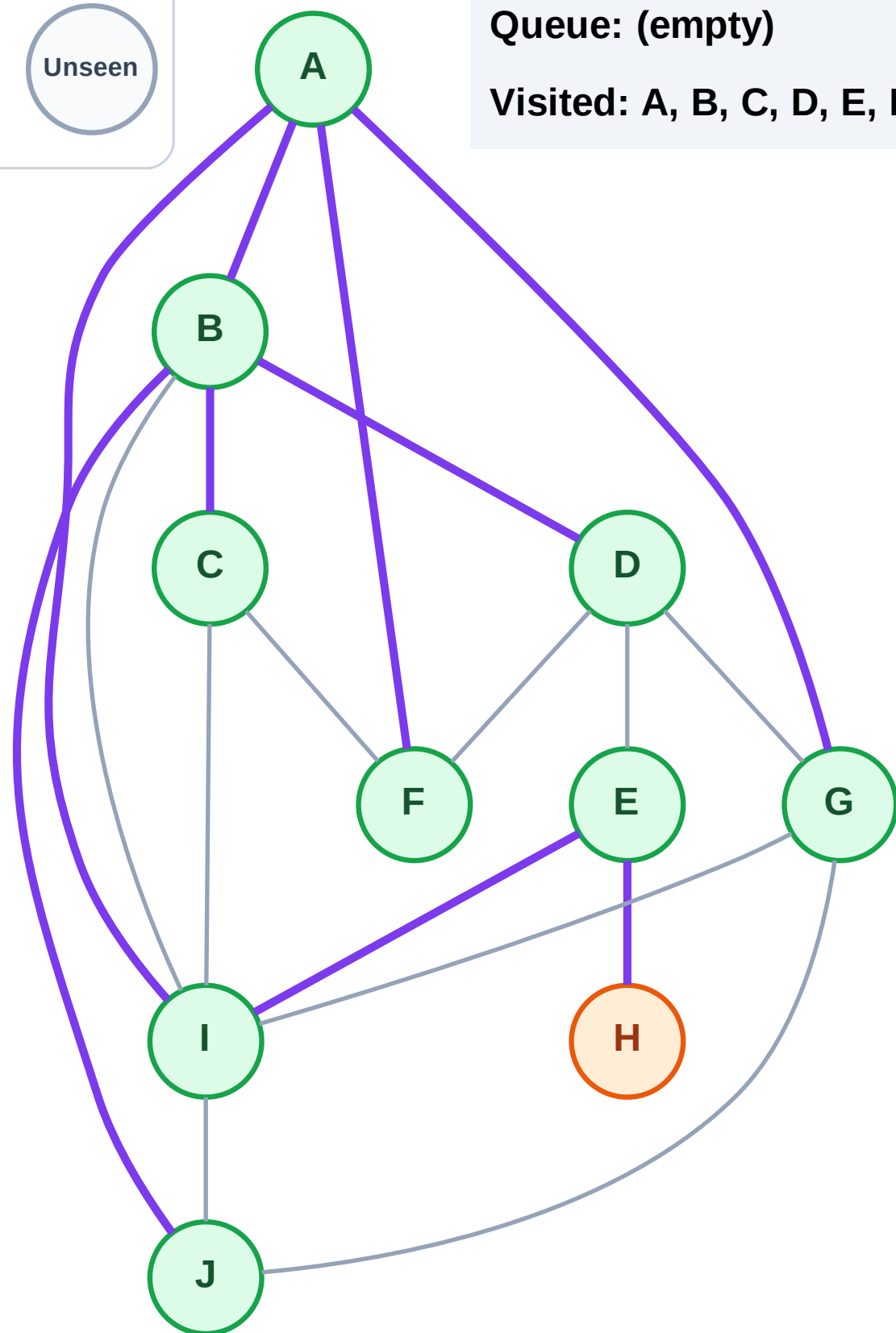
Step 20: Process node H

## Legend



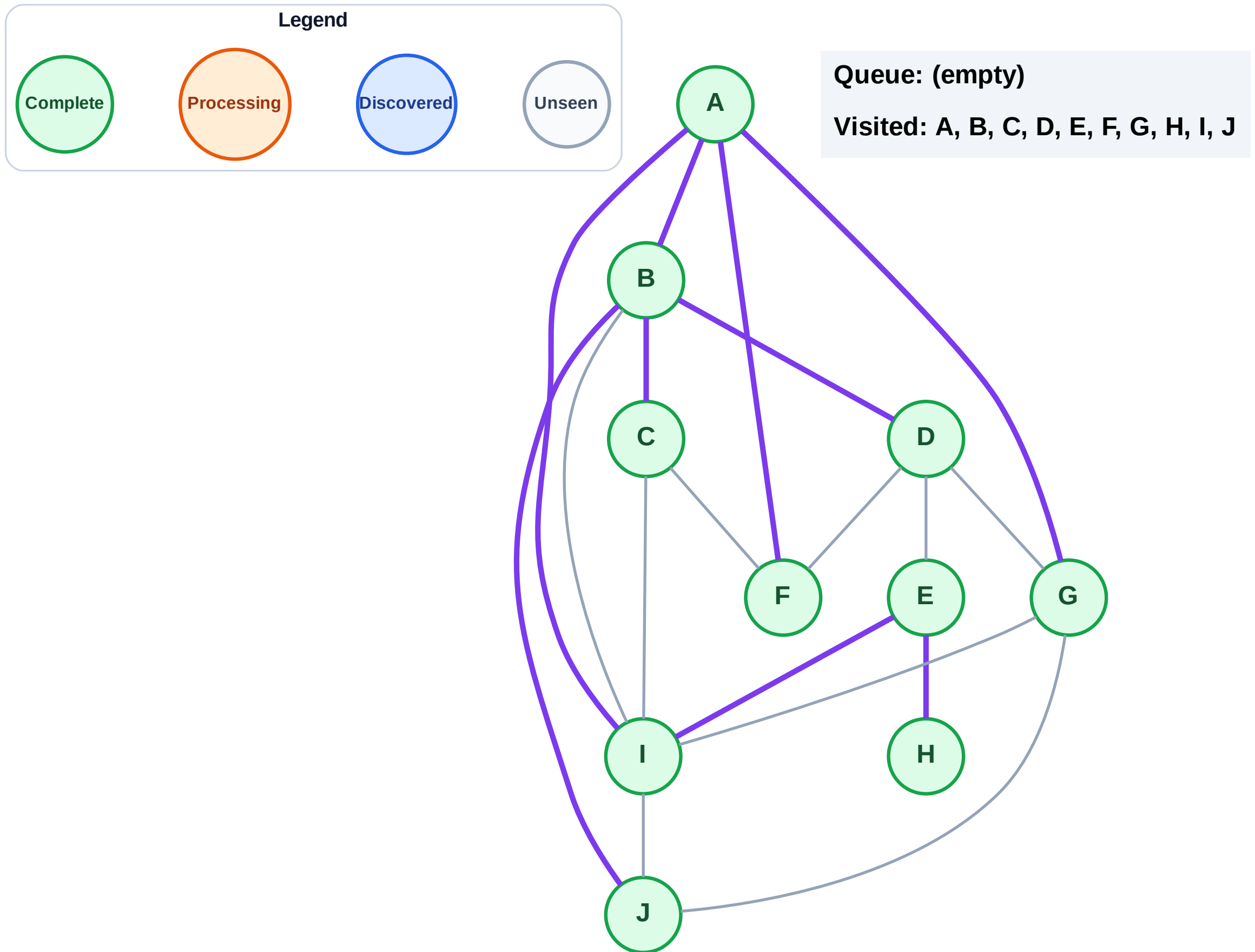
Queue: (empty)

Visited: A, B, C, D, E, F, G, I, J



Step 21: No new neighbors from H

Step 21: No new neighbors from H



# DFS Traversal

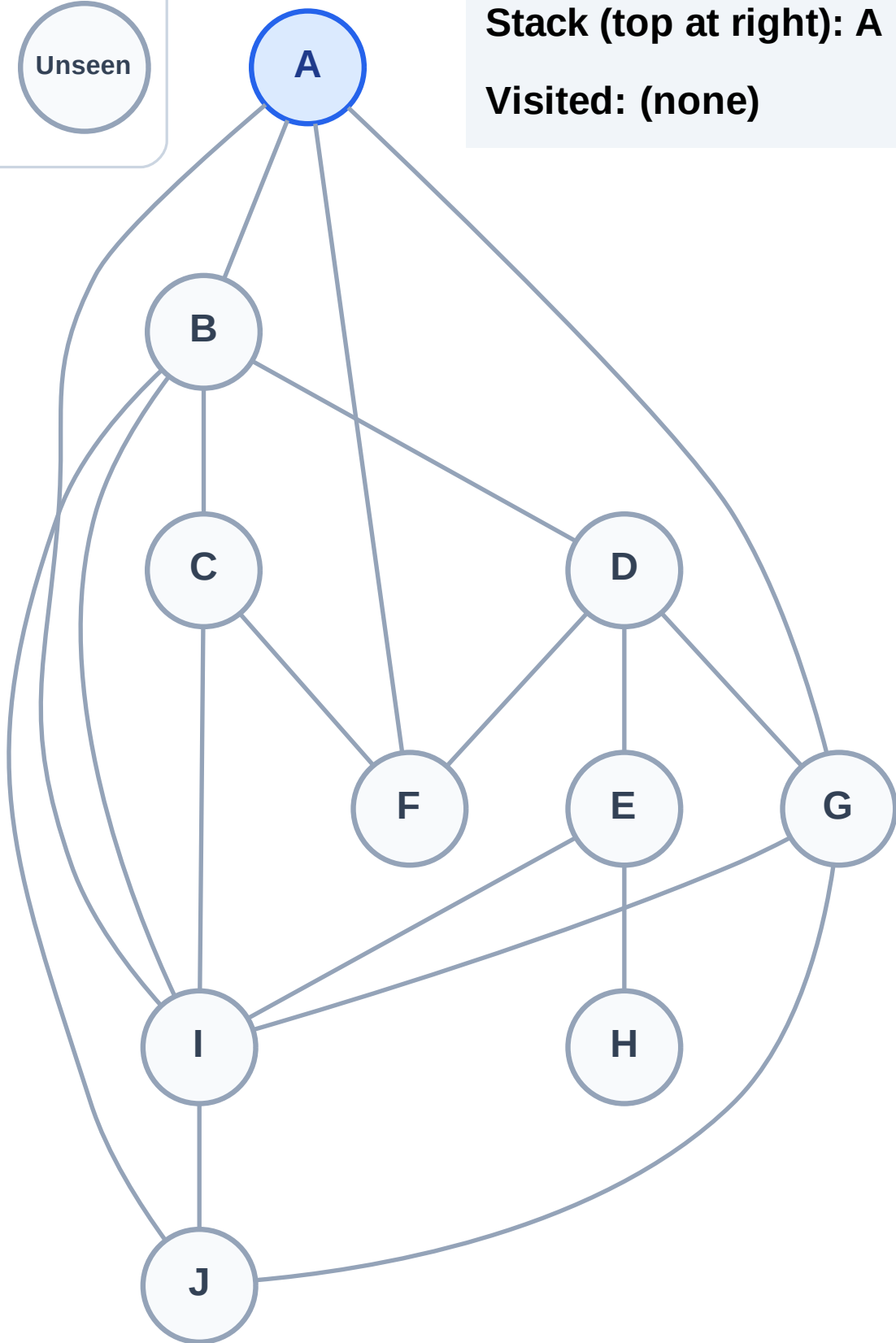
Step 1: Start at A

## Legend



Stack (top at right): A

Visited: (none)



# DFS Traversal

Step 2: Process node A

Legend

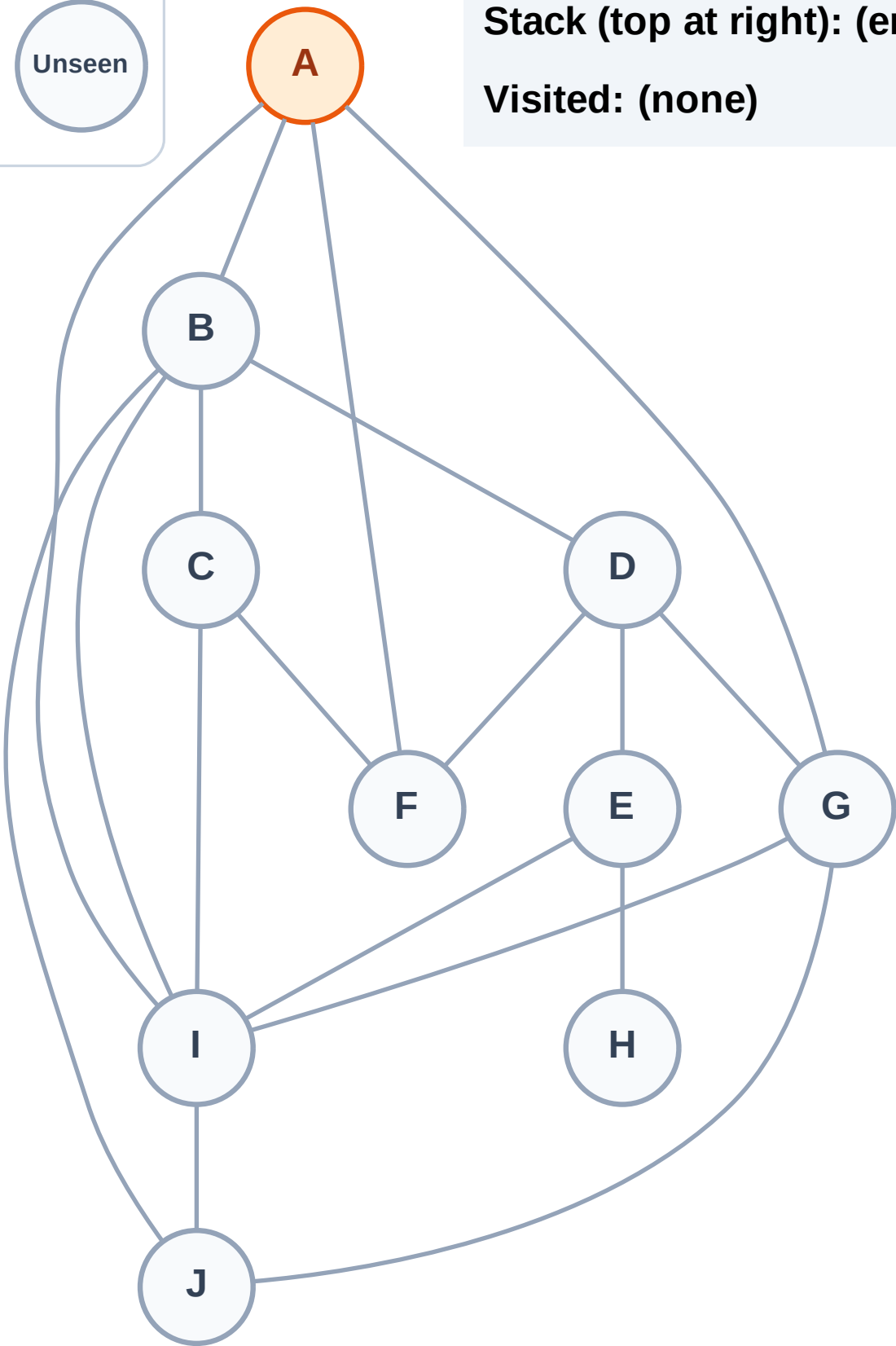
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)  
Visited: (none)





# DFS Traversal

Step 3: Discover I, G, F, B from A

Legend

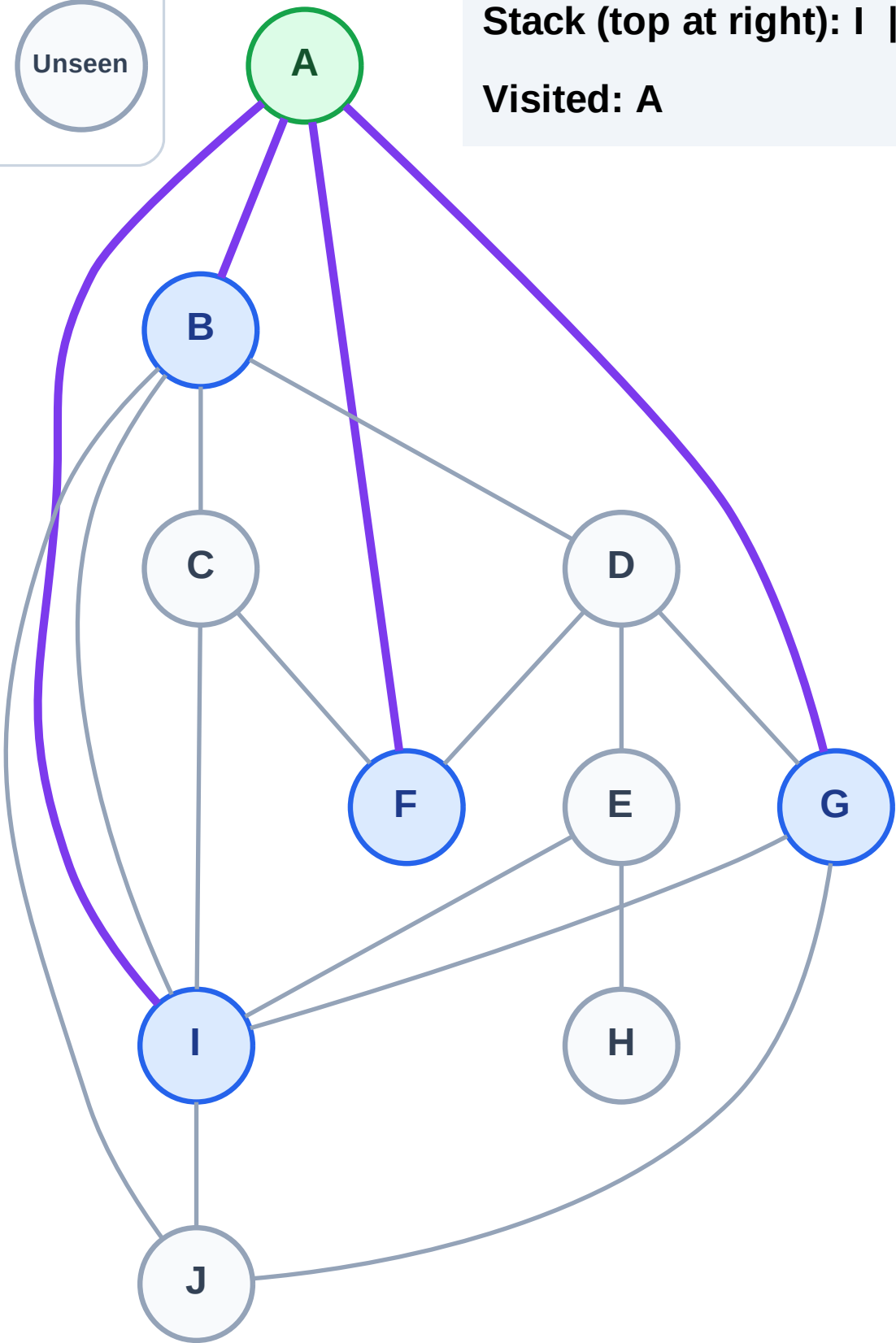
Complete

Processing

Discovered

Unseen

Stack (top at right): I | G | F | B  
Visited: A

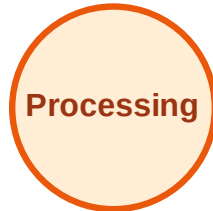




# DFS Traversal

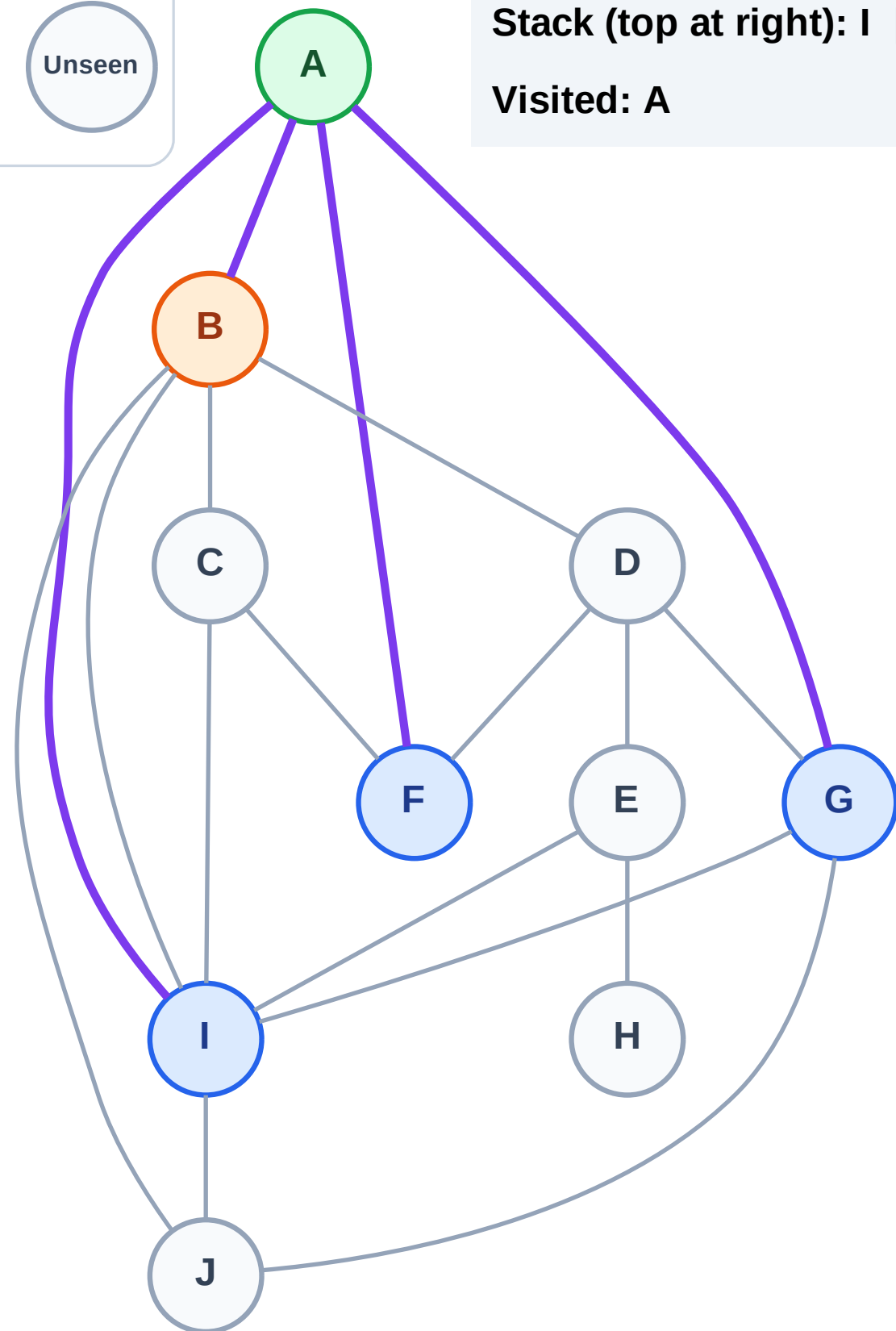
Step 4: Process node B

## Legend



Stack (top at right): I | G | F

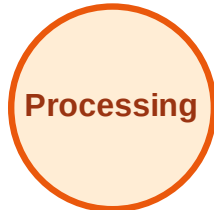
Visited: A



# DFS Traversal

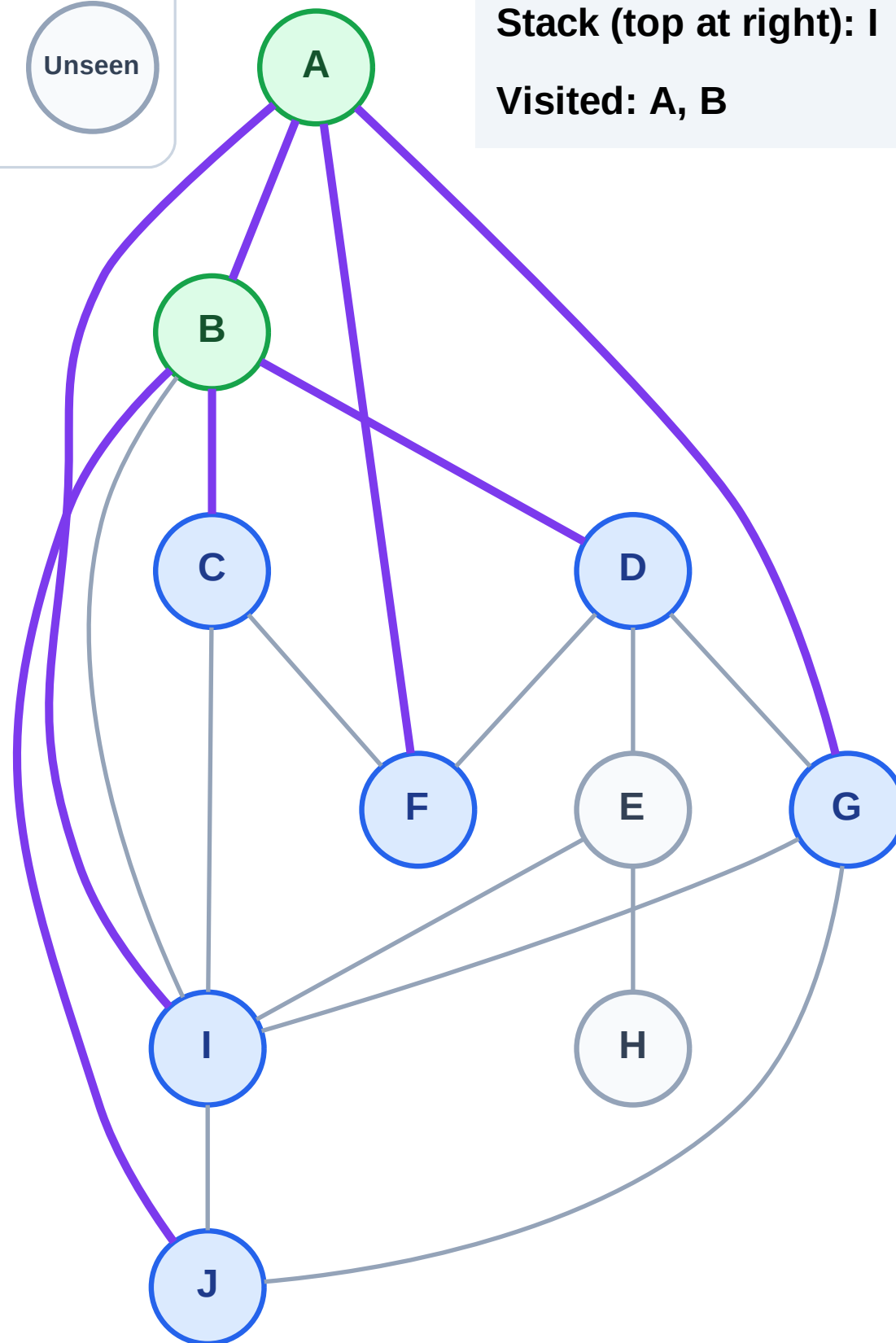
Step 5: Discover J, D, C from B

## Legend



Stack (top at right): I | G | F | J | D | C

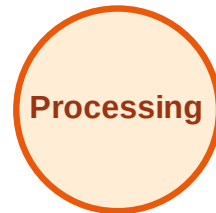
Visited: A, B



# DFS Traversal

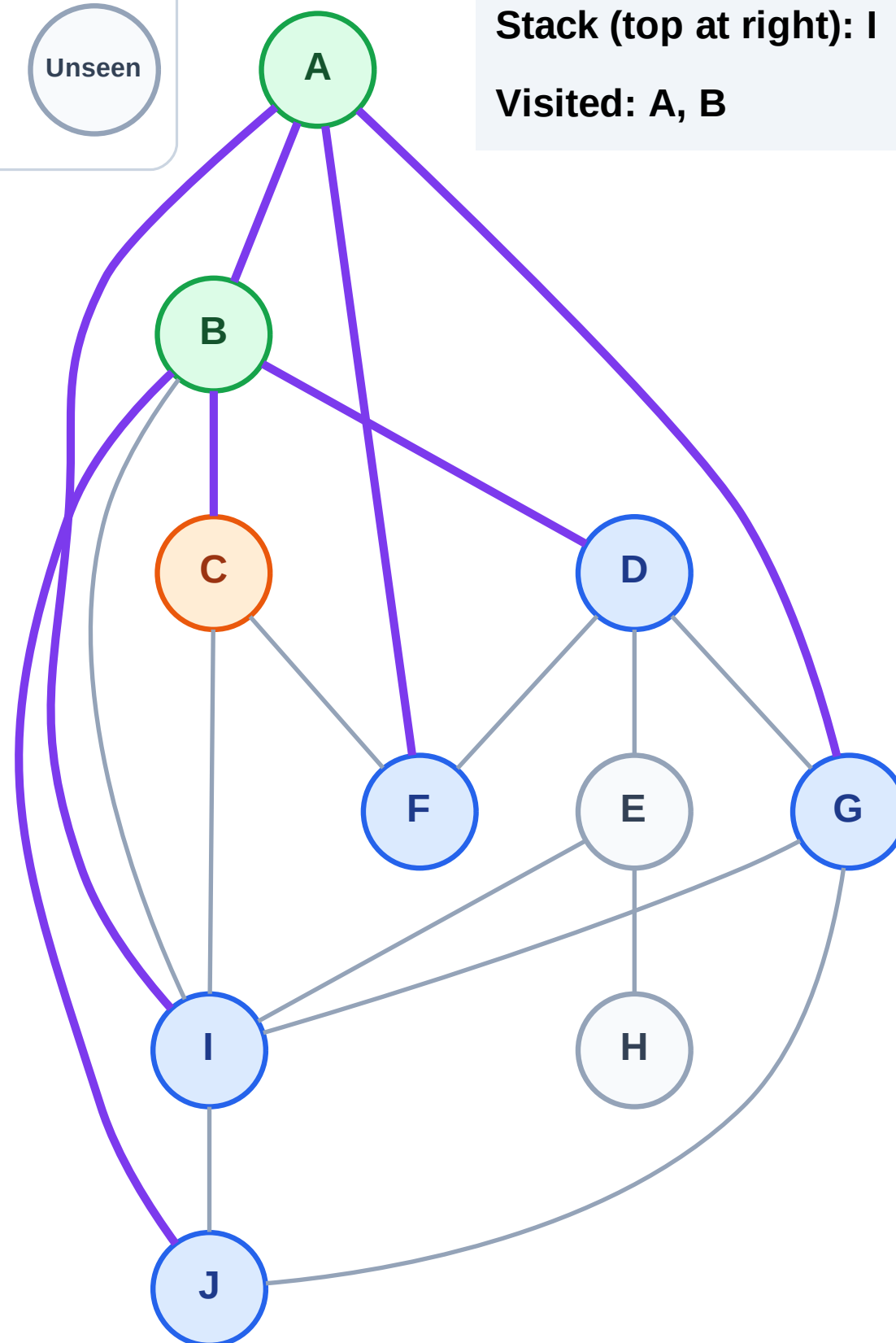
Step 6: Process node C

## Legend



Stack (top at right): I | G | F | J | D

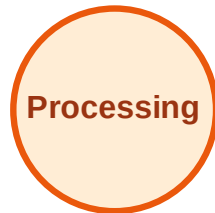
Visited: A, B



# DFS Traversal

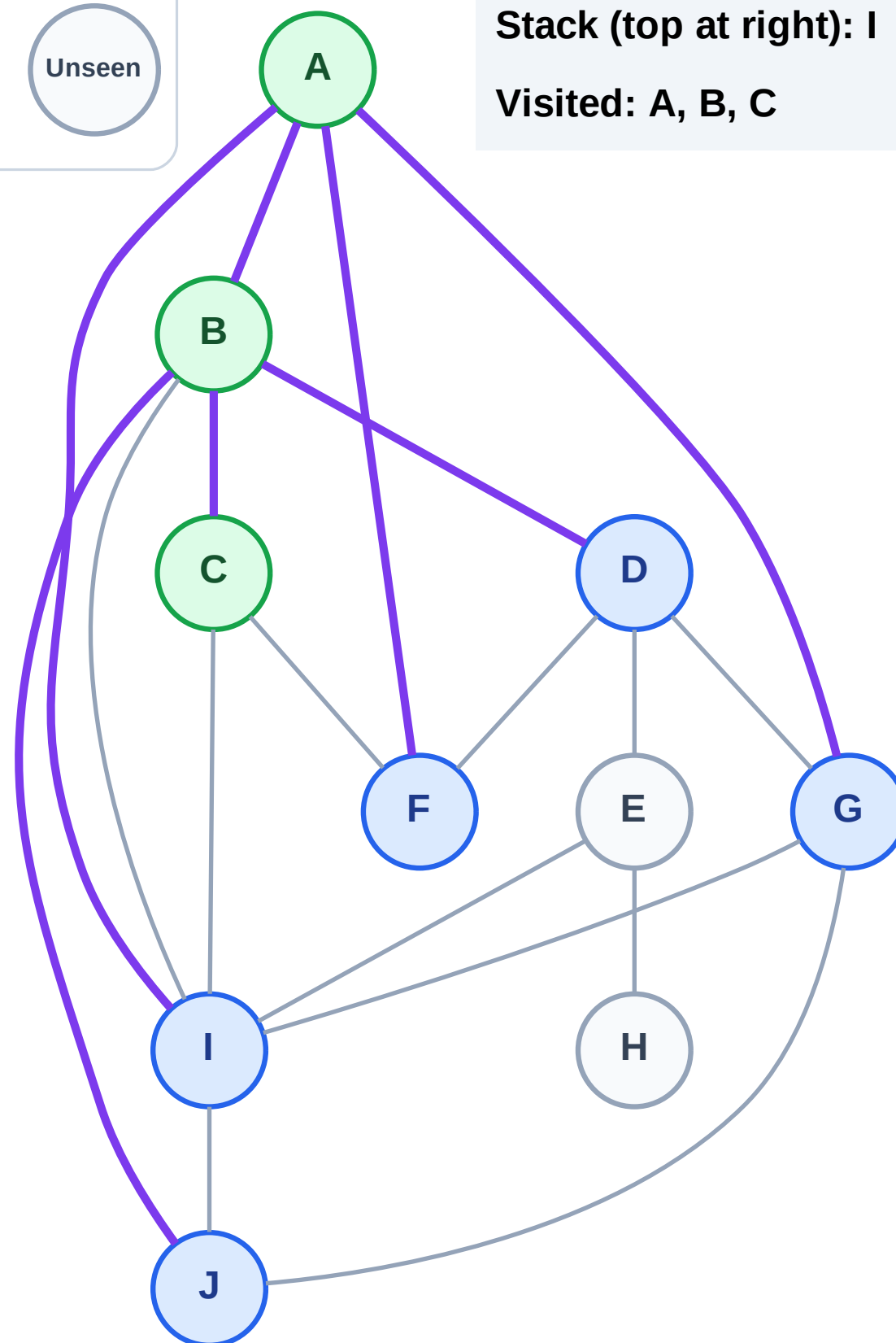
Step 7: No new neighbors from C

## Legend



Stack (top at right): I | G | F | J | D

Visited: A, B, C



# DFS Traversal

Step 8: Process node D

Legend

Complete

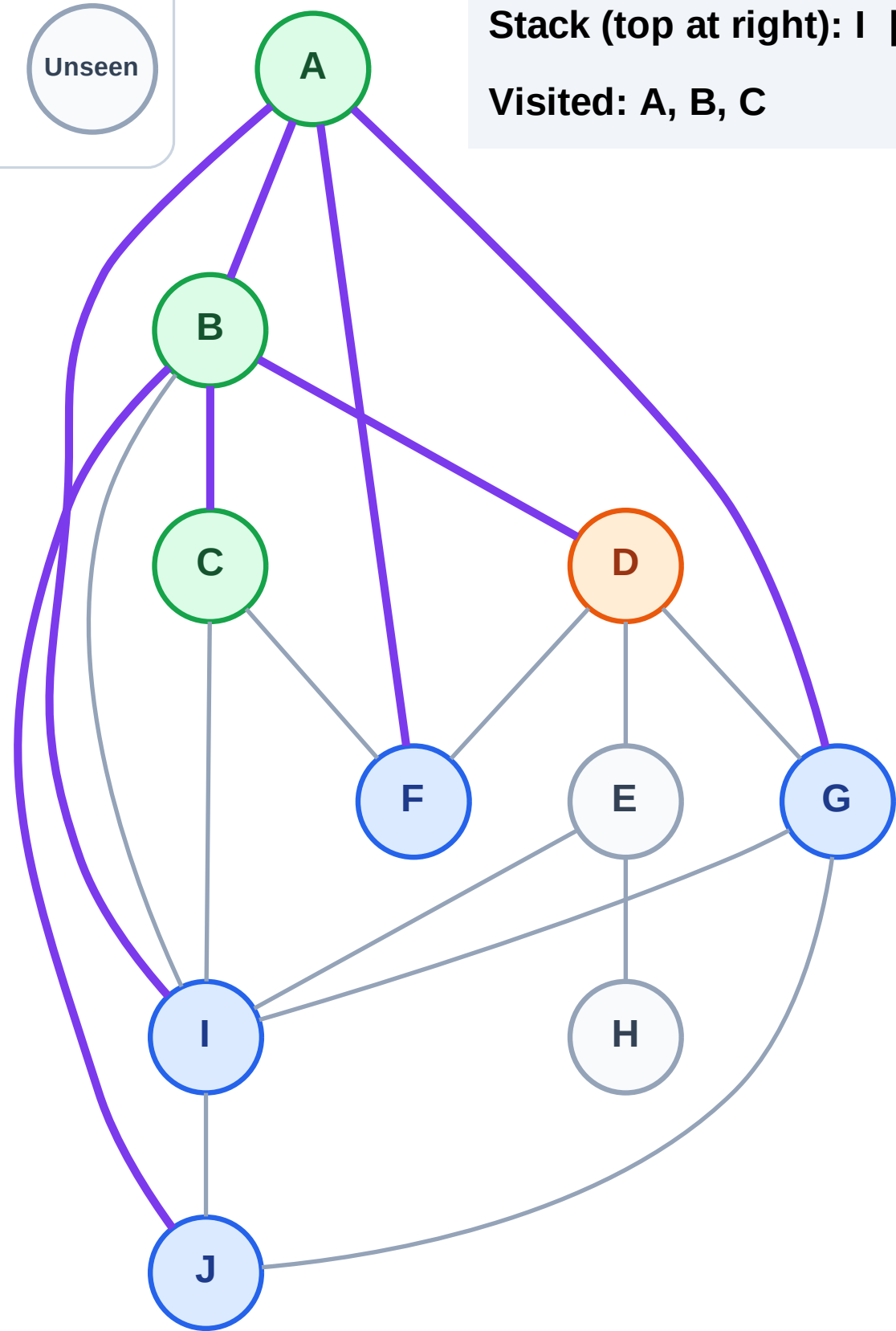
Processing

Discovered

Unseen

Stack (top at right): I | G | F | J

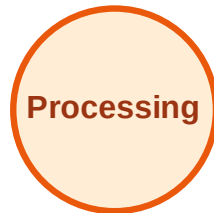
Visited: A, B, C



# DFS Traversal

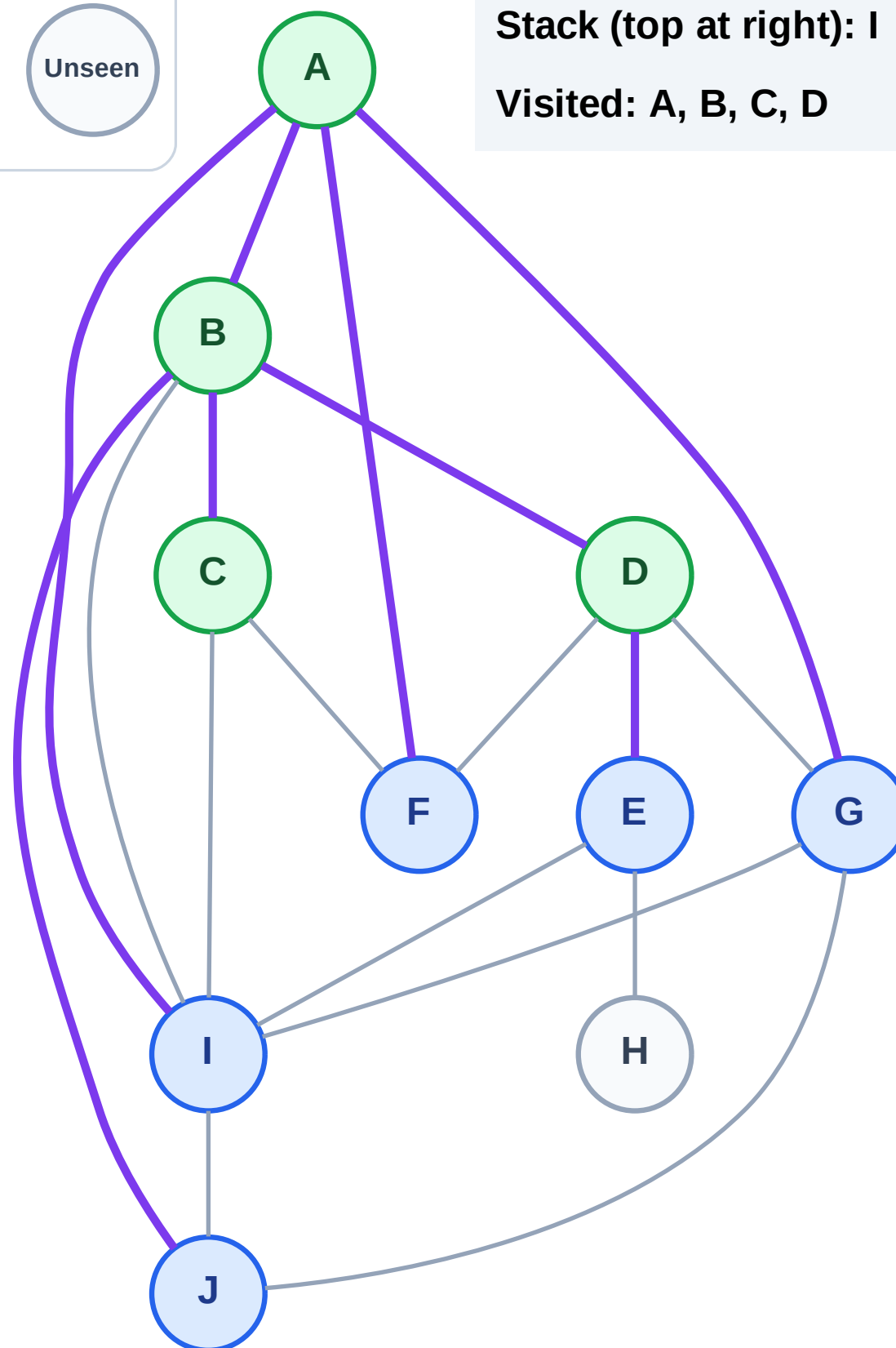
Step 9: Discover E from D

## Legend



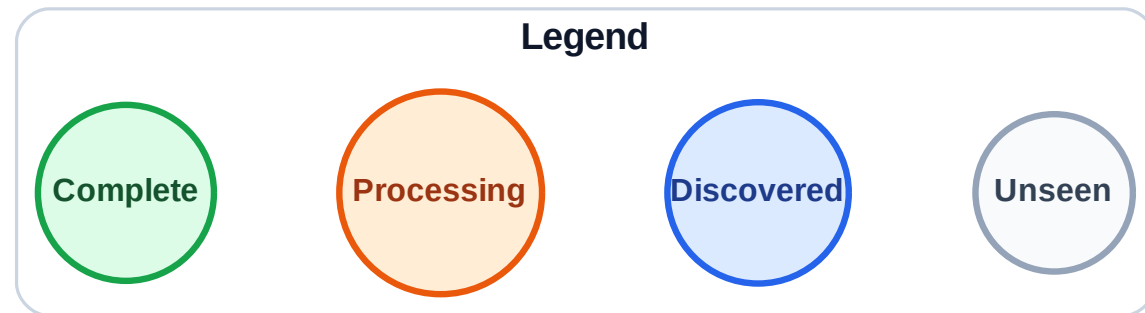
Stack (top at right): I | G | F | J | E

Visited: A, B, C, D

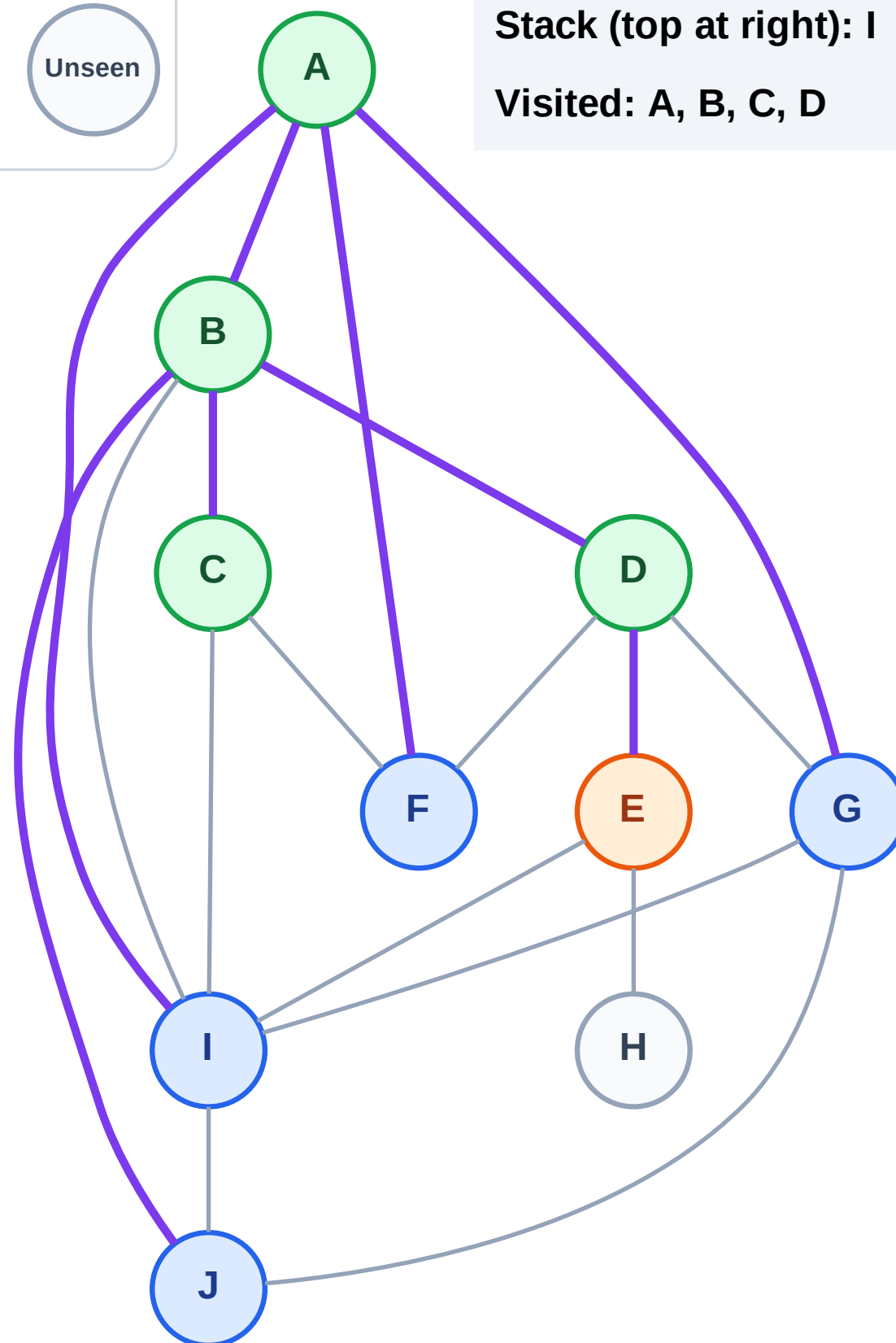


# DFS Traversal

Step 10: Process node E



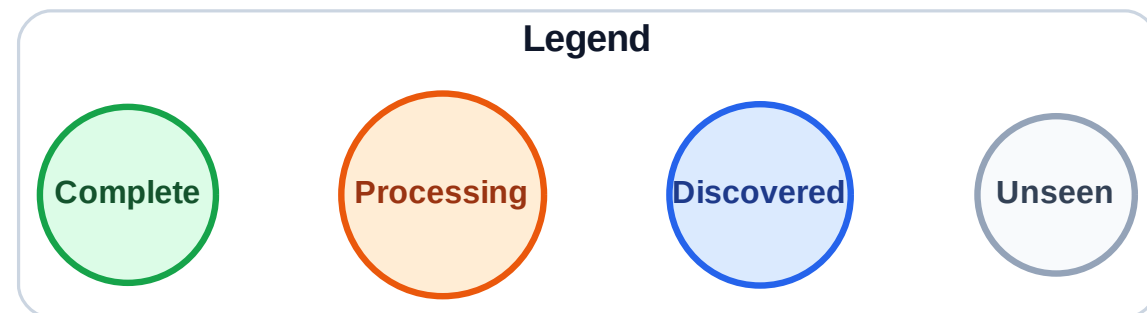
Stack (top at right): I | G | F | J  
Visited: A, B, C, D



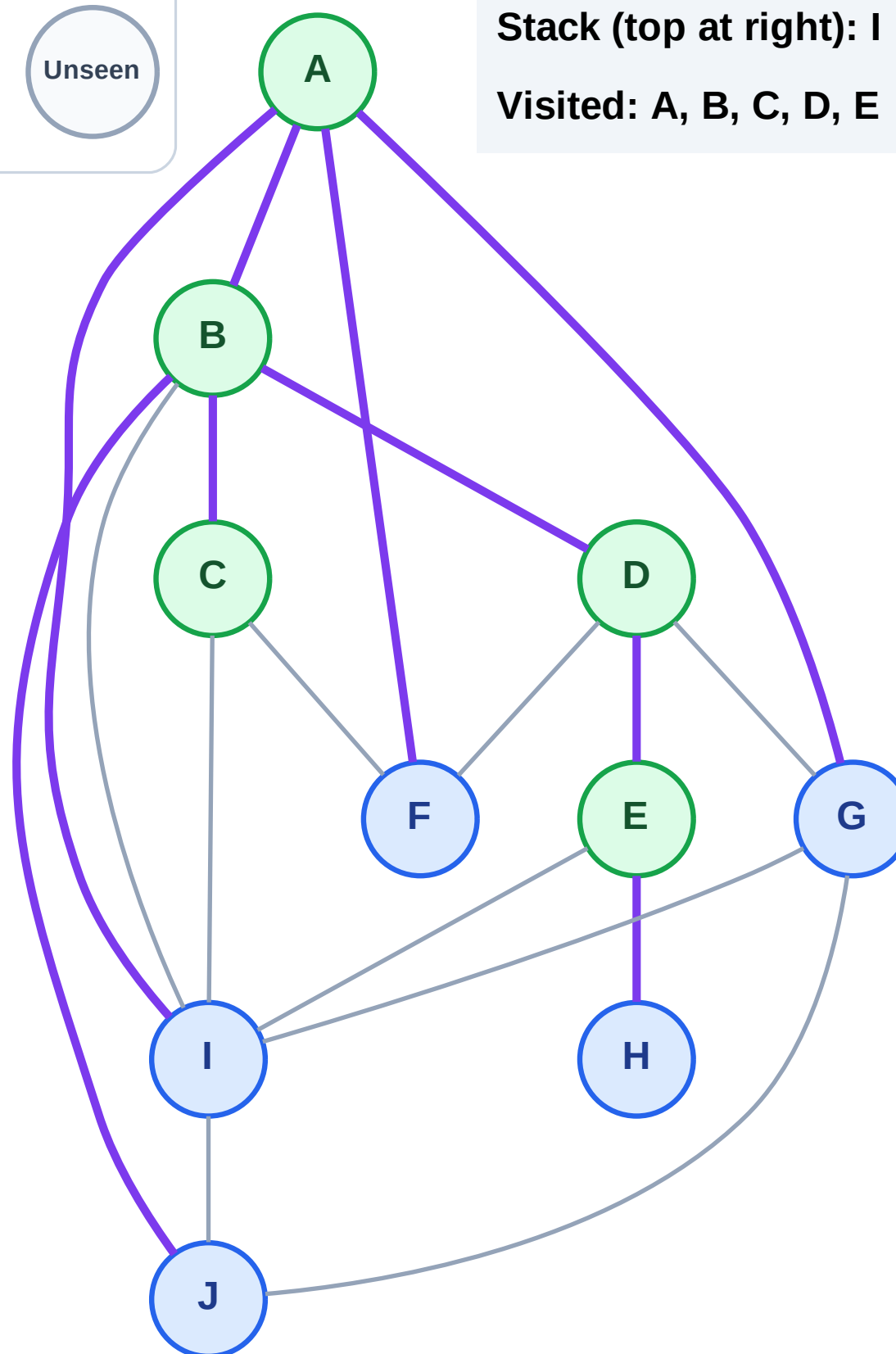


# DFS Traversal

Step 11: Discover H from E



Stack (top at right): I | G | F | J | H  
Visited: A, B, C, D, E

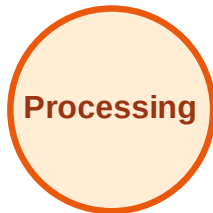




# DFS Traversal

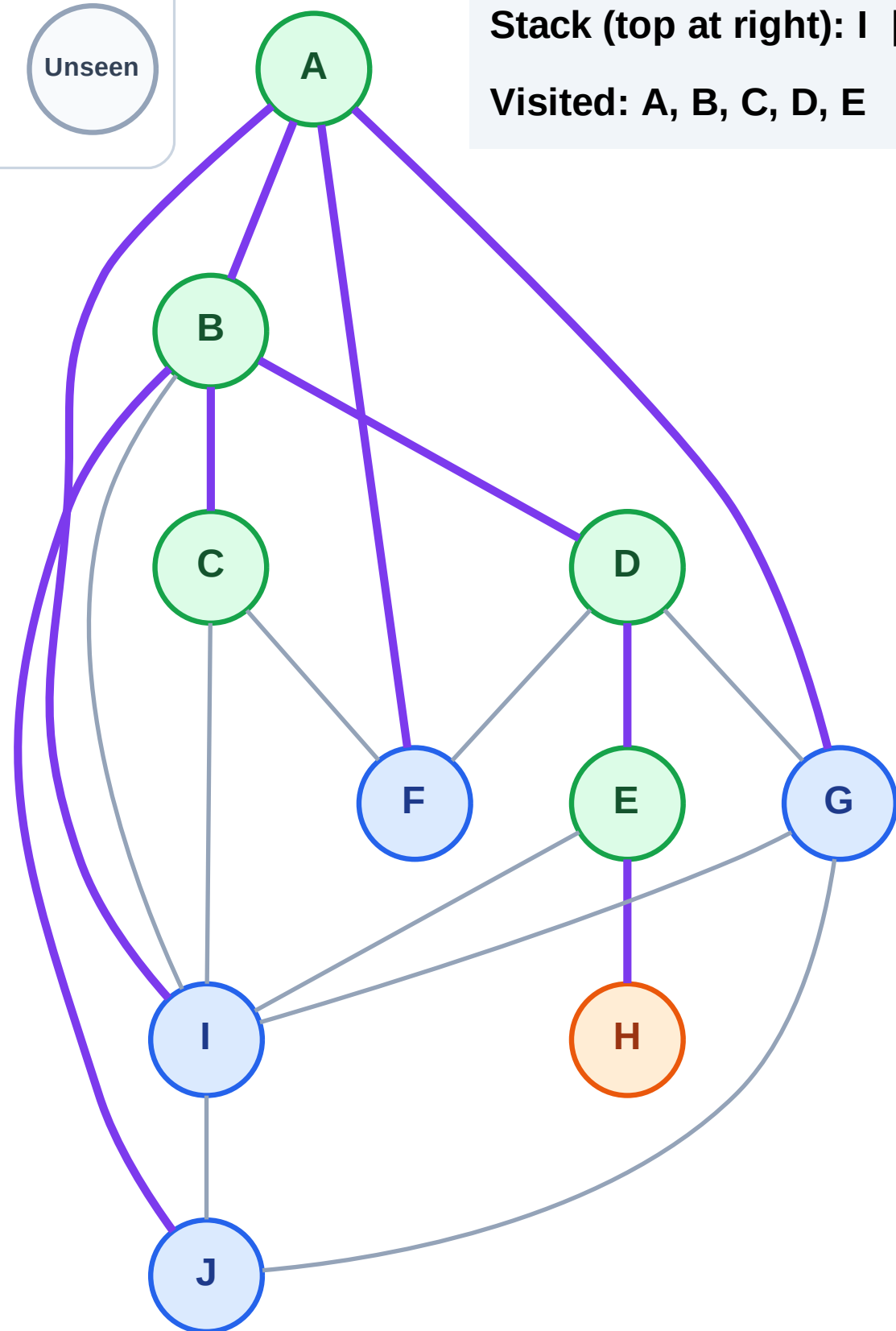
Step 12: Process node H

## Legend



Stack (top at right): I | G | F | J

Visited: A, B, C, D, E



# DFS Traversal

Step 13: No new neighbors from H

Legend

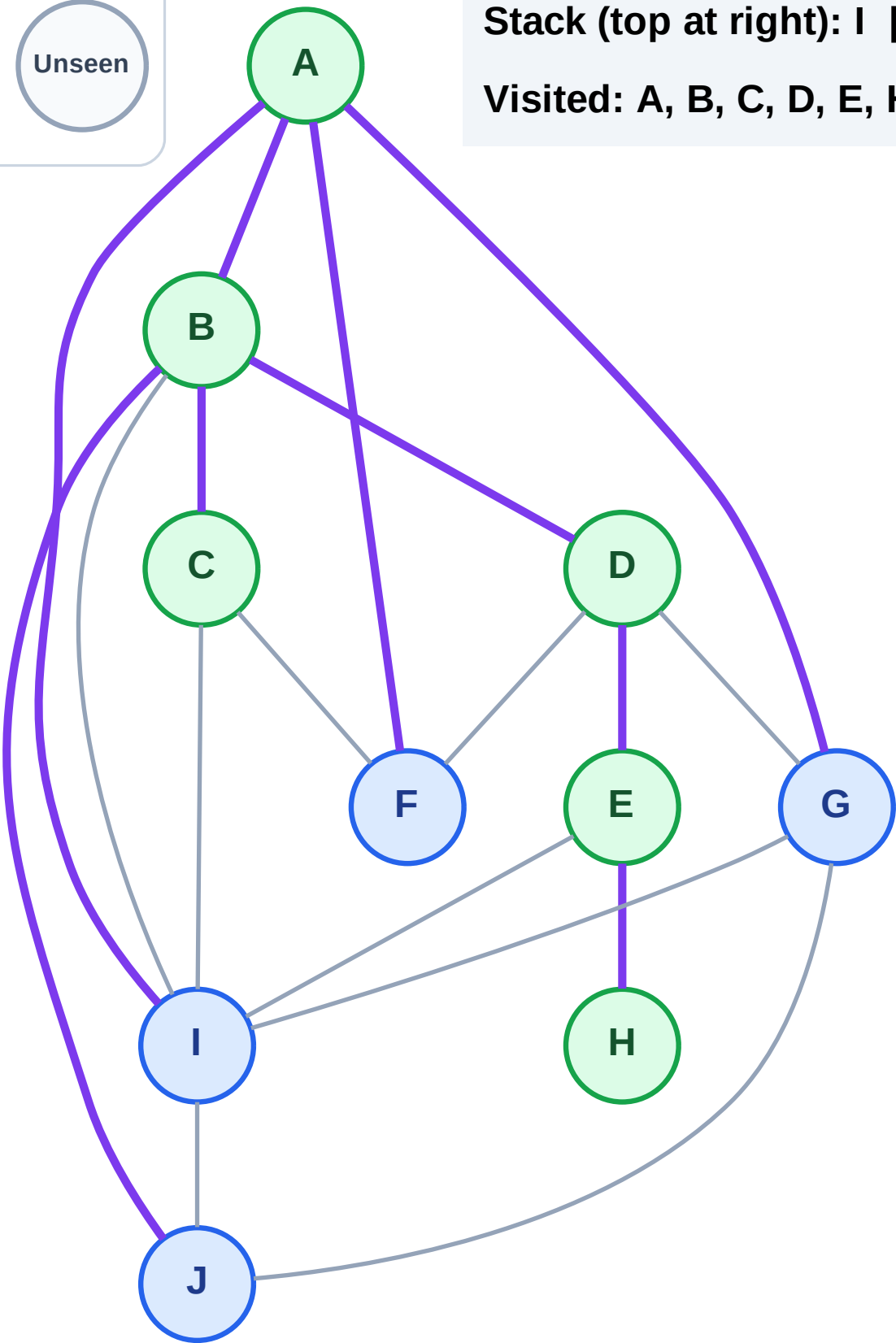
Complete

Processing

Discovered

Unseen

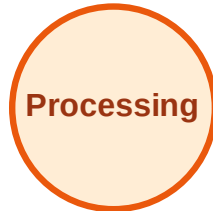
Stack (top at right): I | G | F | J  
Visited: A, B, C, D, E, H



# DFS Traversal

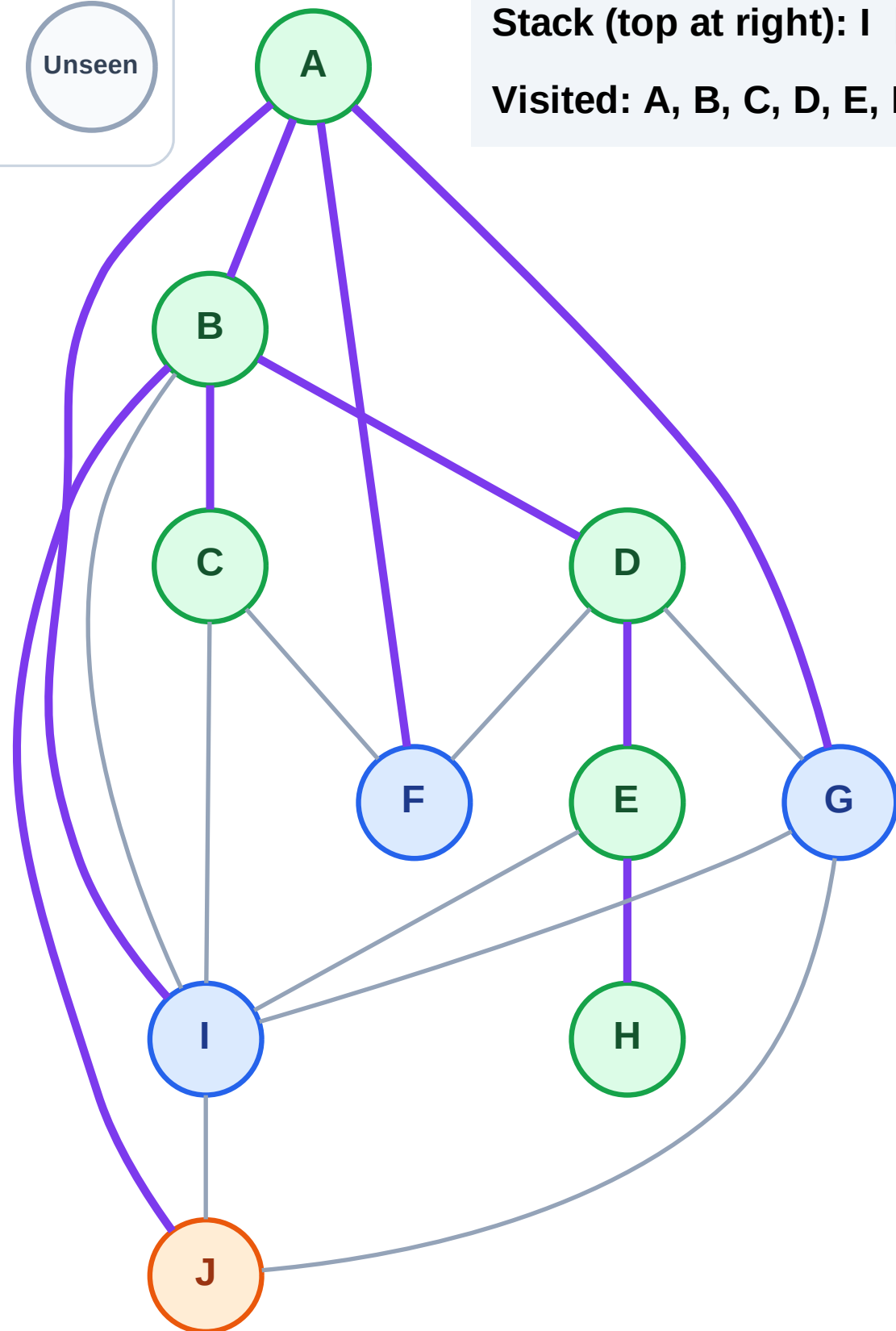
Step 14: Process node J

## Legend



Stack (top at right): I | G | F

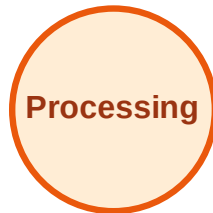
Visited: A, B, C, D, E, H



# DFS Traversal

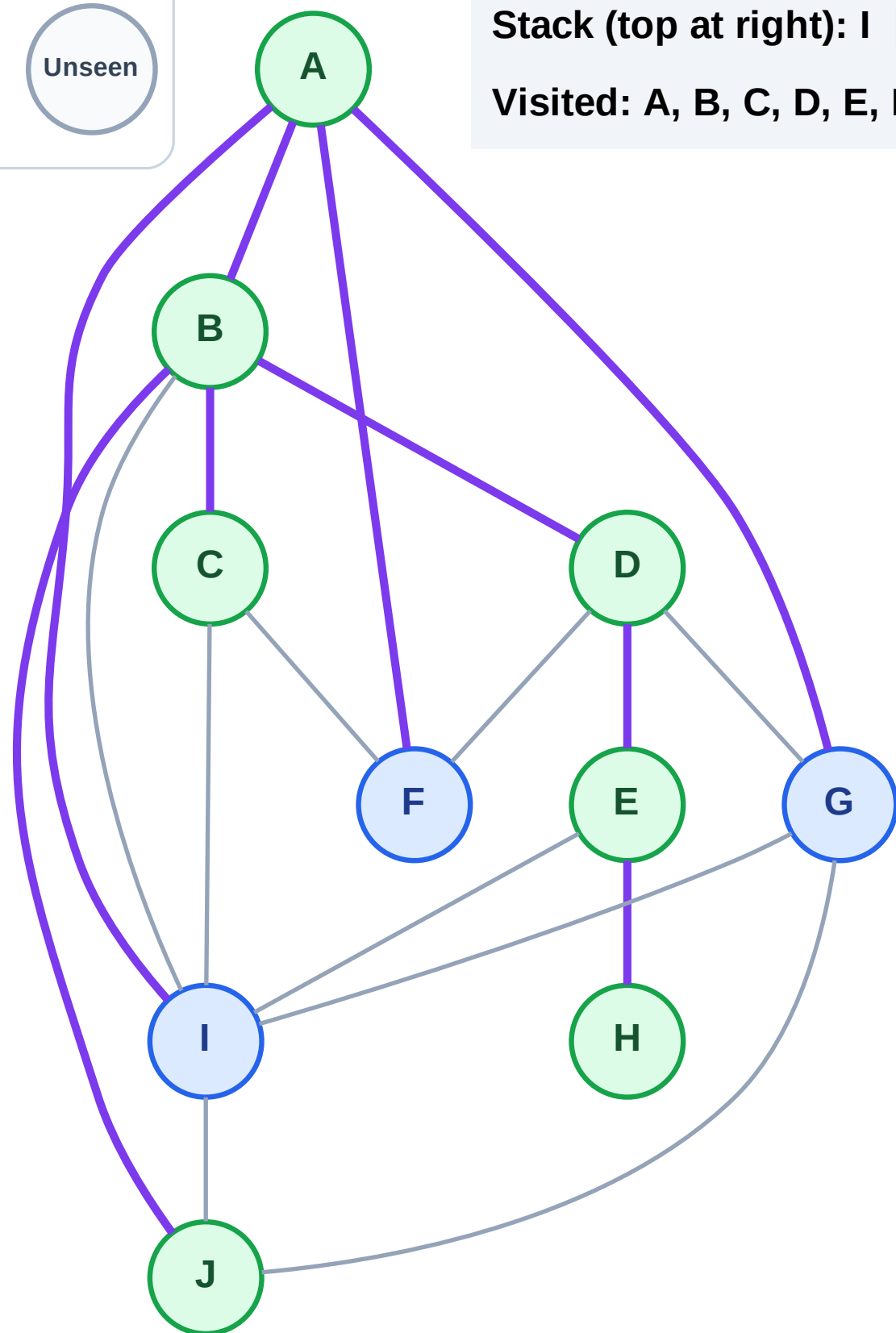
Step 15: No new neighbors from J

## Legend



Stack (top at right): I | G | F

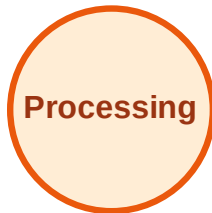
Visited: A, B, C, D, E, H, J



# DFS Traversal

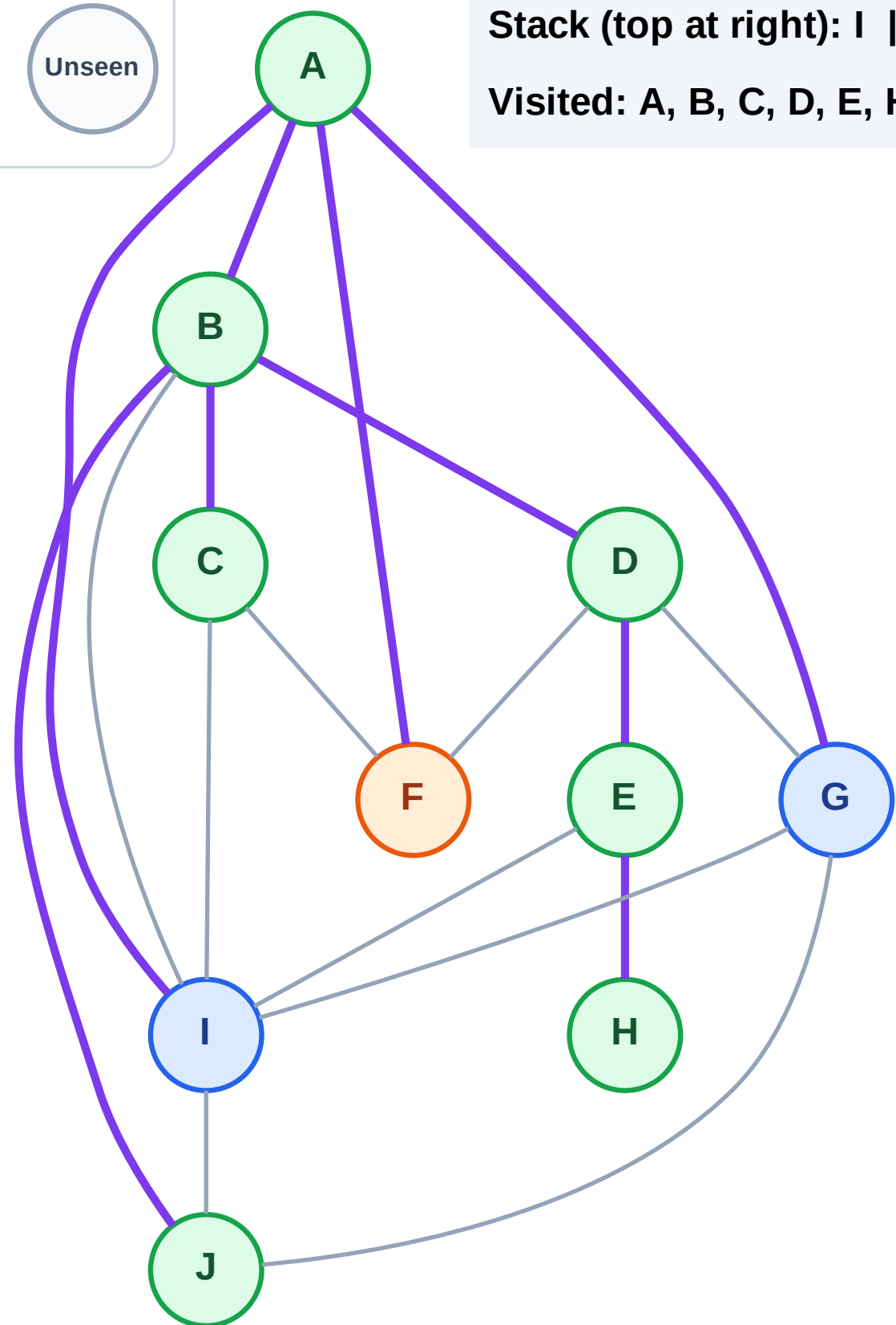
Step 16: Process node F

## Legend



Stack (top at right): I | G

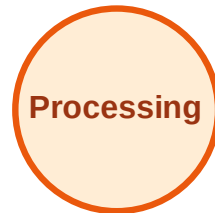
Visited: A, B, C, D, E, H, J



# DFS Traversal

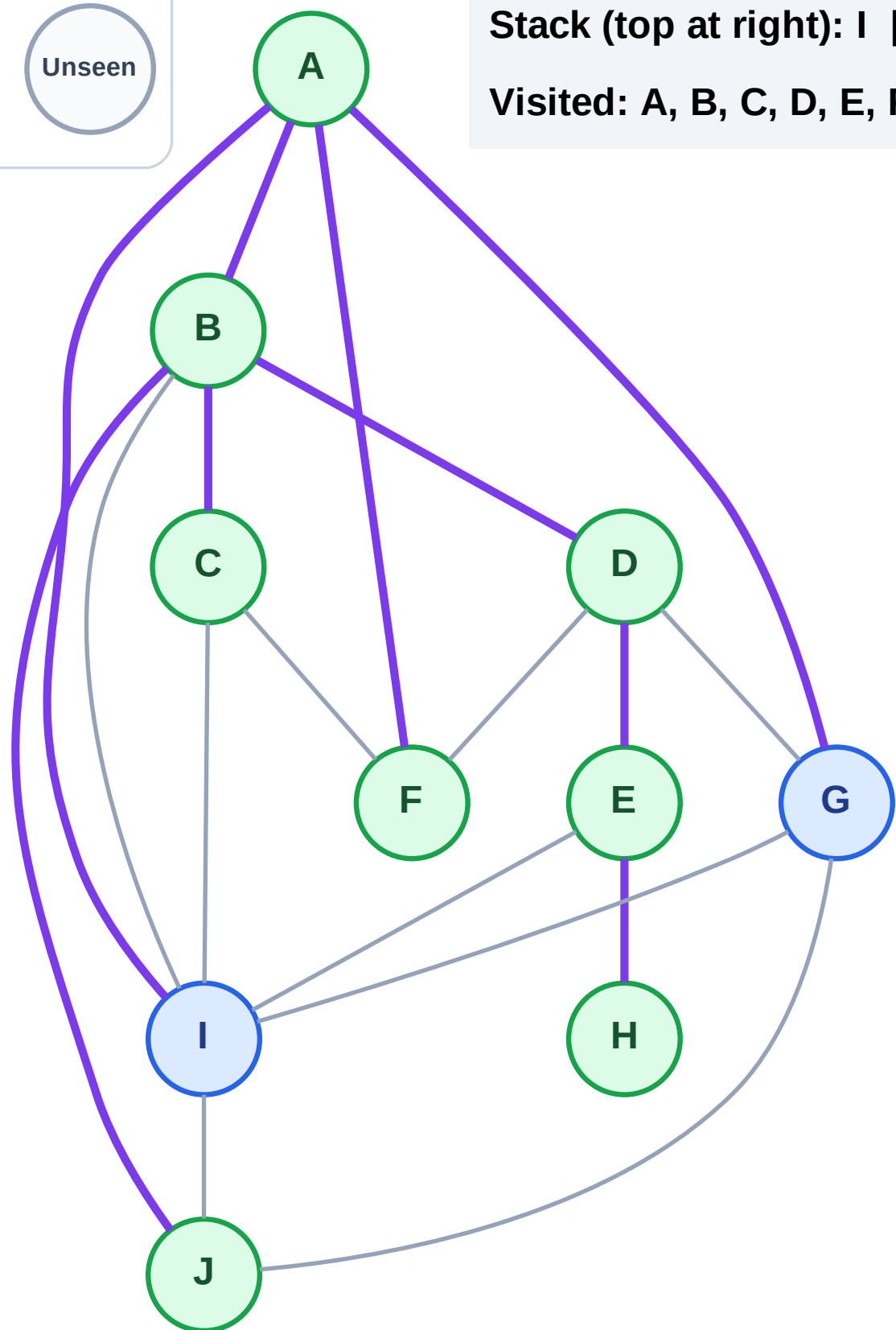
Step 17: No new neighbors from F

## Legend



Stack (top at right): I | G

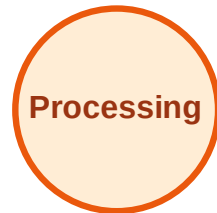
Visited: A, B, C, D, E, F, H, J



# DFS Traversal

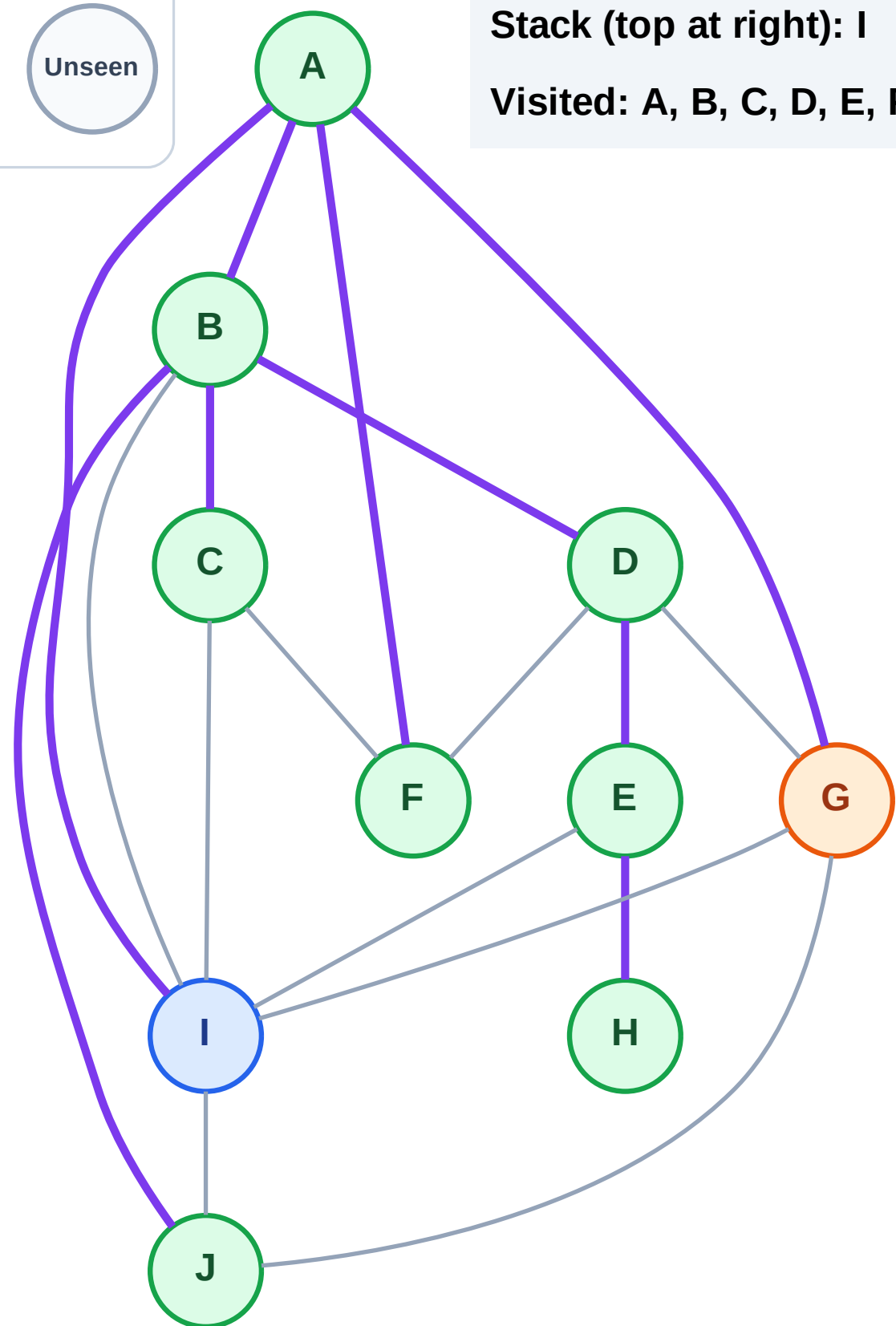
Step 18: Process node G

## Legend



Stack (top at right): I

Visited: A, B, C, D, E, F, H, J

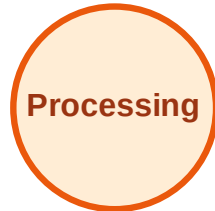




# DFS Traversal

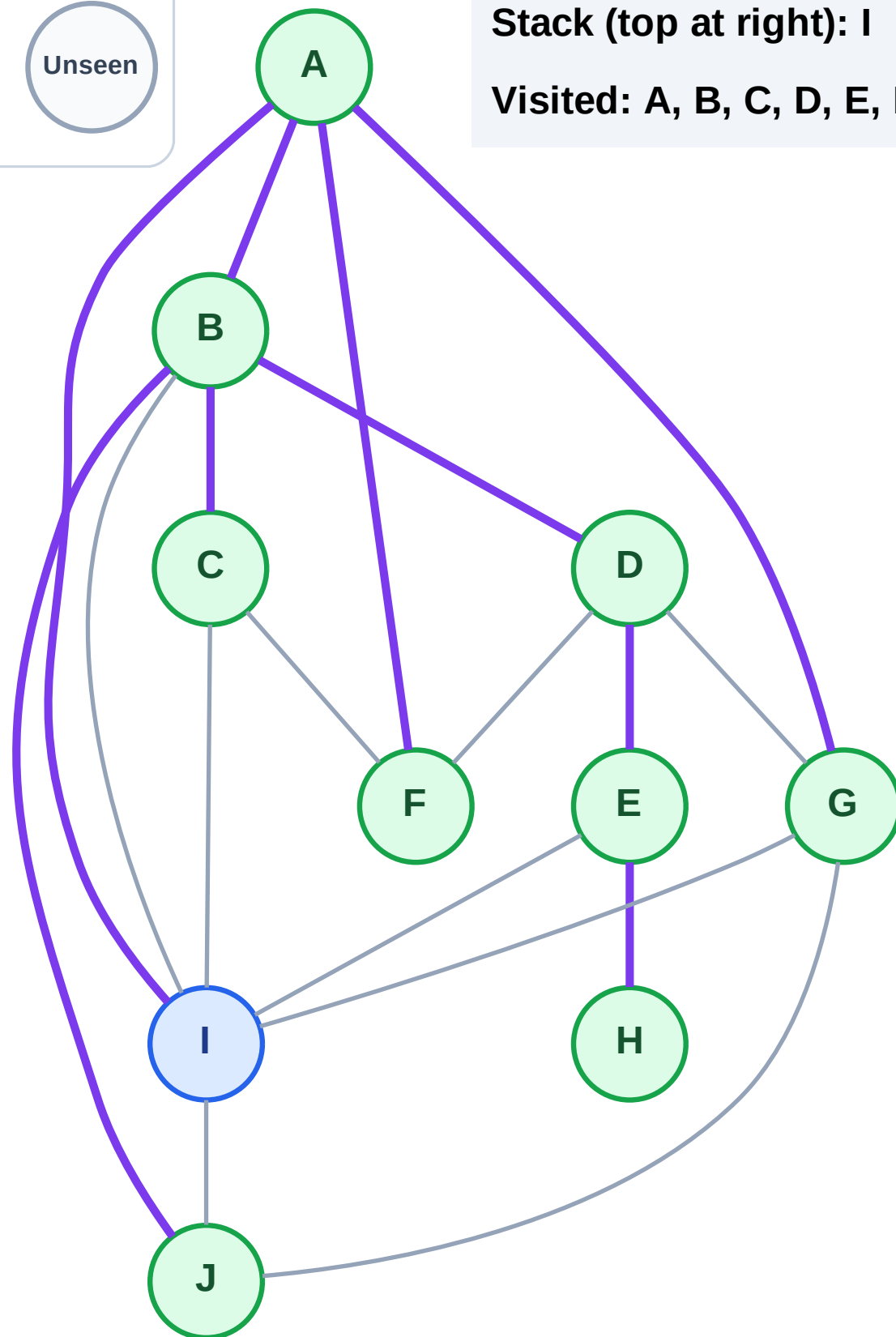
Step 19: No new neighbors from G

## Legend



Stack (top at right): I

Visited: A, B, C, D, E, F, G, H, J

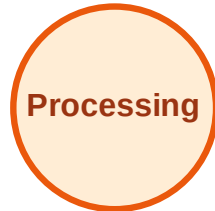




# DFS Traversal

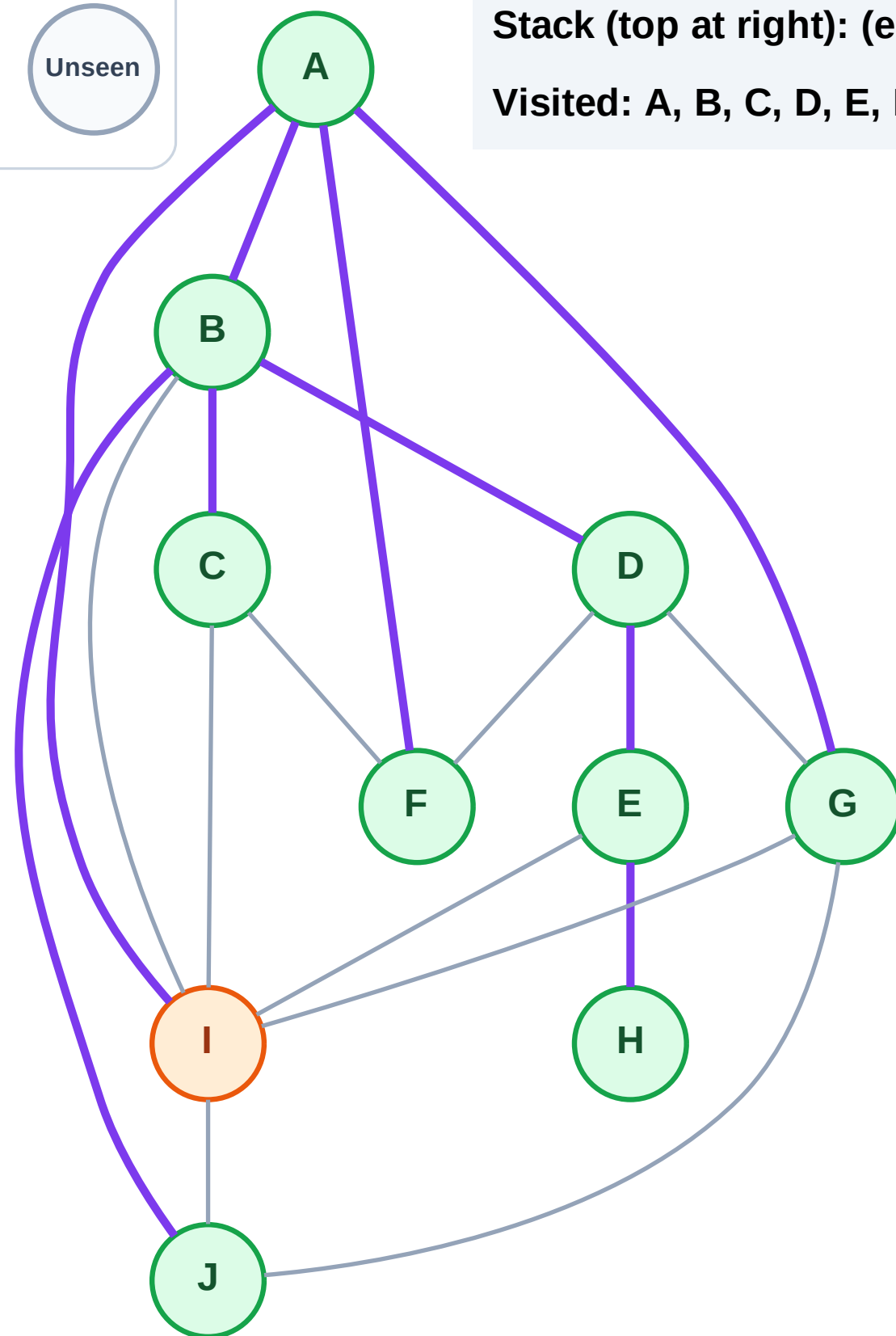
Step 20: Process node I

## Legend



Stack (top at right): (empty)

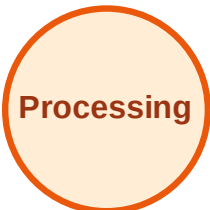
Visited: A, B, C, D, E, F, G, H, J



# DFS Traversal

Step 21: No new neighbors from I

## Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

